

Exploratory Methods for Spatio-Temporal Data

Spatio-temporal processes typically involve dependencies across a large range of spatial and temporal scales. In order to understand such processes, scientific theories are developed and observations are taken. These spatio-temporal data are a window to the underlying process; our statistical goals are to visualize, summarize, and infer (including spatio-temporal prediction) the process' behavior within and between relevant scales of variability. Exploratory-data-analytic methods of visualization and summarization are considered in this chapter. The goal is to use these exploratory methods to suggest relationships that can be incorporated into spatio-temporal models for the purpose of inference (e.g., Section 9.1).

A key challenge in spatio-temporal data analysis is the extraction of information from potentially large spatio-temporal datasets generated by data-assimilation algorithms, deterministic models, observation networks, or remote-sensing platforms. There are a number of approaches that have proven to be helpful in exploratory data analysis and diagnostics for model building, when working with spatio-temporal data. In this chapter, we review them in some detail, occasionally extending them, and we provide applications for which they can be used. We note that some of these methods have been developed in subject-matter disciplines (primarily meteorology, climatology, and oceanography) where statisticians have not generally played a prominent role.

We realize that a static medium (e.g., pages in this book) can be limiting for data that are inherently dynamical, and we look forward to seeing e-books in the future with new media for exploratory methods (e.g., hyperlinks that allow presentation of animation). The development of exploratory and diagnostic methods for spatio-temporal data is an important research topic and will remain so in the future.

5.1 VISUALIZATION

As with any statistical data analysis, we should always try to visualize (summaries of) the data. There are particular challenges in visualizing spatio-temporal data, since there are typically a minimum of three dimensions (two-dimensional space $\times$ one-dimensional time) to consider simultaneously. Of course, there are as many visualization approaches as there are data analysts, so we must be selective. In this section, we describe several simple visualization tools that can be quite informative when applied in practice.

5.1.1 Animations

Perhaps the most useful visualization for spatio-temporal datasets with two or three spatial dimensions is to use the special nature of the temporal dimension to sequence the data's variability and show it as an animation, or "movie." In cases where the data are completely gridded in space and evenly spaced in time, the production of animations is relatively simple. In fact, some computational packages (e.g., Matlab®) have built-in tools to produce animations.

In cases where there are missing observations or where observation times are unequally spaced, a simple interpolation (in space and time) algorithm to fill in missing observations for the purposes of animation (in time) might be applied. Alternatively, for two-dimensional spatial data that are not on a regular lattice, animations can be created from scatterplots. For example, at any snapshot in time, observations can be represented by the size and/or color of circles at each spatial location (e.g., see Figure 5.5 below). By exploiting the unidirectionality of the time dimension, these types of maps can be sequenced into an animation. As mentioned in our introductory remarks, an e-book version would allow animations to be hyperlinked. Animations could also be made available at selected websites (e.g., Section 9.1 presents forecasts for sea surface temperature in the tropical Pacific Ocean that have been animated; see SSES, 2009).

5.1.2 Marginal and Conditional Plots

Space(1-D)/Time Plots

A common method of illustrating wave propagation in the geophysical sciences is by use of image plots showing one-dimensional (1-D) space (typically either latitude or longitude) on the x -axis and time on the y -axis, where time progresses as one looks from top to bottom of the plot. For example, if the x -axis represents longitude from west to east, then coherent patterns going down the plot and to the right suggest propagation eastward, and patterns slanting to the left down the page are indicative of propagation westward. These are called *Hovmöller diagrams* in the atmospheric-sciences literature (Hovmöller, 1949).

An illustration of a space/time plot is given in Figure 5.1, where monthly Pacific sea surface temperature (SST) anomalies (averaged from 1° South latitude to 1° North latitude) are plotted as a function of longitude (x -axis) for

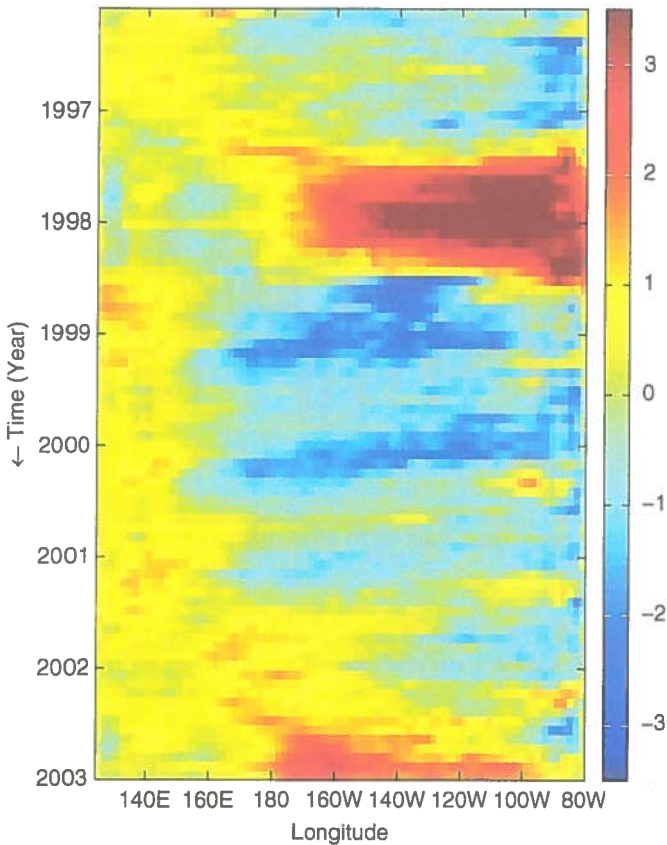


Figure 5.1 Sea surface temperature anomalies ($^{\circ}\text{C}$) averaged between latitudes 1°S and 1°N . The y axis refers to time in monthly increments (increasing down the page from 1996 to 2003), and the x axis corresponds to longitude in 2° increments (throughout the equatorial Pacific region).

the period from January 1996 to December 2002. The dark-red anomalies correspond to a major El Niño event, and the dark-blue anomalies correspond to La Niña events. Notice that there is evidence of propagation in this figure. For example, both La Niña events show clear propagation from the eastern portion of the domain into the western portion through time. There is also visual evidence to suggest that there is some precursor warming around 180° East (and West) longitude, preceding the warm (El Niño) events.

Time-Series Plots

It is always helpful to plot time series associated with various spatial locations or regions of space. For example, consider the tropical Pacific SST anomalies discussed in the previous example. There are several regions in the tropical Pacific that are deemed representative of the El Niño and La Niña phenomena. One of these, the so-called “Niño 3” region, corresponds to an average of SSTs

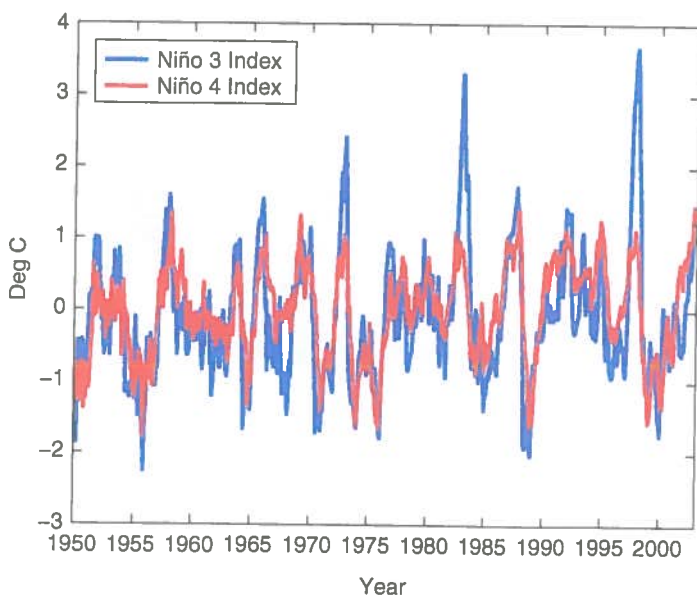


Figure 5.2 Monthly Niño 3 and Niño 4 indices from January 1950 to December 2002. The y axis shows anomalies in $^{\circ}\text{C}$ and the x axis shows time in monthly increments.

over the region 5°N to 5°S and 150°W to 90°W . Another representative region, the “Niño 4” region, is an average of SSTs over 5°N to 5°S and 160°E to 150°W . Time series of these aggregates from January 1950 to December 2002 are shown in Figure 5.2. Clearly, these two time series are highly correlated, but they illustrate that the spatial regions they represent do not show identical warming and cooling.

It is often useful to consider aggregating over the entire spatial domain. Here we use as an illustrative example a study of the Eurasian Collared Dove (ECD, *Streptopelia decaocto*) in North America. The ECD was first observed in the USA in Florida during the mid-1980s (Romagosa and Labisky, 2000). Not only has its range increased as it has spread through the southeastern United States, but it has experienced growth in population as well (Section 9.3). Data from the North American Breeding Bird Survey (BBS; Robbins, Bystrak, and Geissler, 1986) are collected at many sampling routes; Figure 5.3 shows an aggregate of the BBS counts for the ECD from 1986 through 2003. One can see from this that the BBS counts have increased exponentially through time, as is characteristic for a recently introduced invasive species.

Spatial Maps

Just as it is useful to look at time series for individual spatial locations or for spatial regions, it is also useful to look at spatial maps for given times, sequences of times, or aggregates over time. Returning to the tropical Pacific

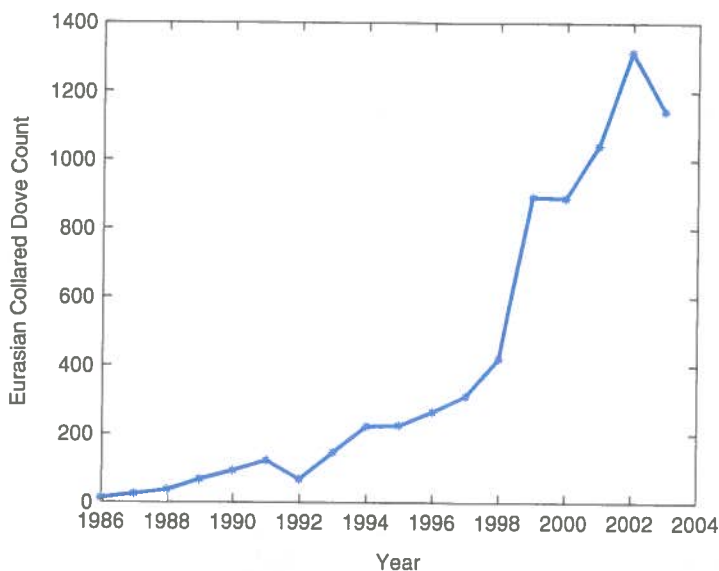


Figure 5.3 Yearly North American Breeding Bird Survey counts of Eurasian Collared Doves from 1986 to 2003. The y axis shows counts of the Eurasian Collared Dove and the x axis shows time in yearly increments.

SST example, Figure 5.4 shows monthly SST anomalies over the tropical Pacific region from February 1998 through January 1999. The observations are gridded at $2^\circ \times 2^\circ$ lat–lon resolution, and there are $m = 2261$ Pacific-Ocean $2^\circ \times 2^\circ$ pixels in the region of interest (from 30° South to 30° North latitude.) One can immediately see that the sequence of monthly maps shows the rapid transition from a warm event (El Niño) to a cold event (La Niña). This rapid transition from one extreme upper-ocean state to another is characteristic of the El Niño/Southern Oscillation (ENSO) phenomenon. It is also obvious from Figure 5.4 that the spatial pattern of the warm and cold events are not the same. This type of nonlinearity is another characteristic feature of the ENSO phenomenon. In Section 9.1, a BHM is fitted to these spatio-temporal data, and conclusions from this exploratory analysis are confirmed. Inference resulting from the fitted BHM is given by Berliner, Wikle, and Cressie (2000), and subsequent monthly updates can be found in SSES (2009).

The Pacific SST data in Figure 5.4 are on a regular grid. One can also look at a sequence of maps when the data are not on a regular grid, such as for the Eurasian Collared Dove (ECD) data. Figure 5.5 shows a 3-yearly sequence of yearly BBS sampling-route counts for the ECD from 1988 to 2003. In this figure, the ECD's relative abundance is represented by both the size and color of the circles. From this sequence of maps, it is immediately obvious that the range and abundance of the ECD has expanded substantially over the 15 years covered by these data.

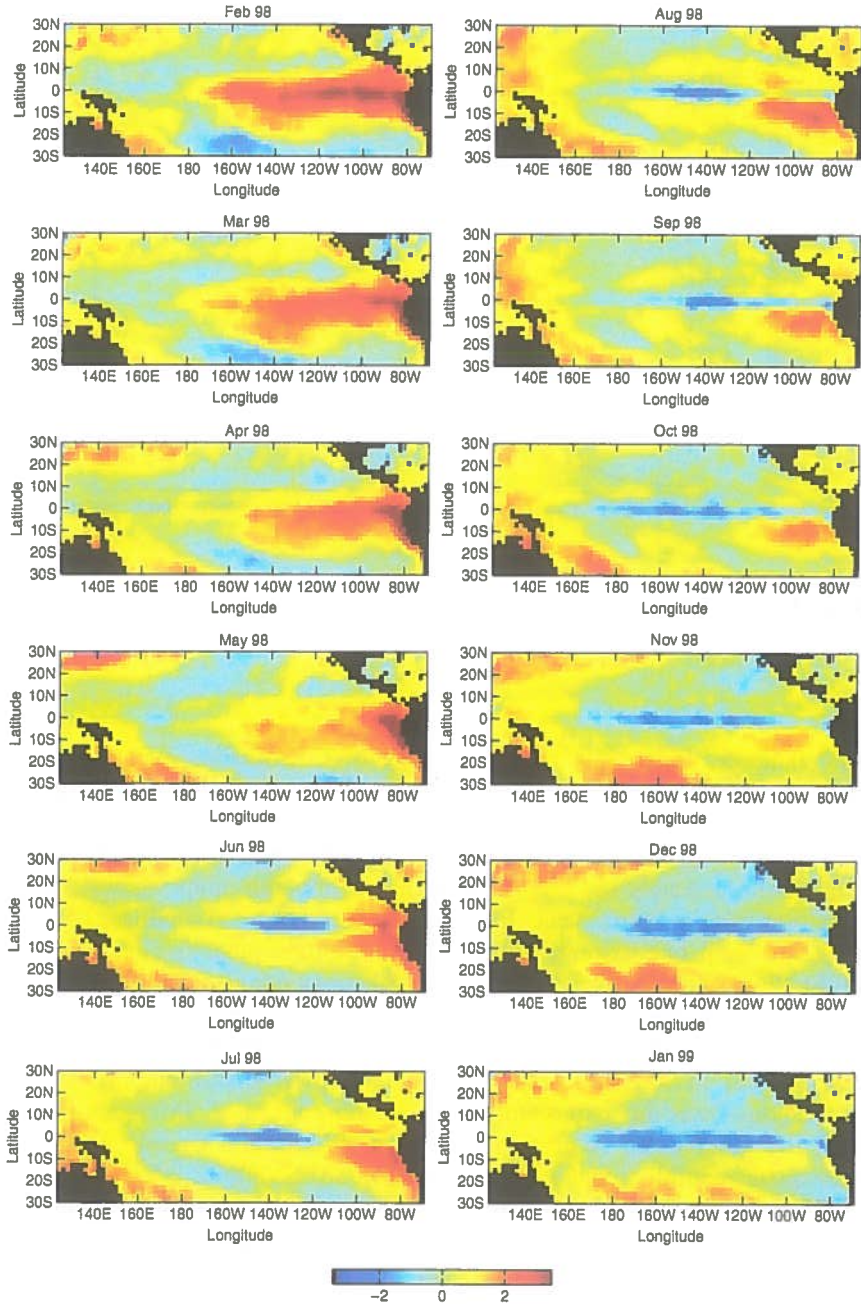


Figure 5.4 Sea surface temperature anomalies in the tropical Pacific region from February 1998 to January 1999. The beginning of this period shows strong warming off the coast of South America (El Niño phenomenon). The end of this period shows the development of an area of cooling in the central Pacific (La Niña phenomenon).

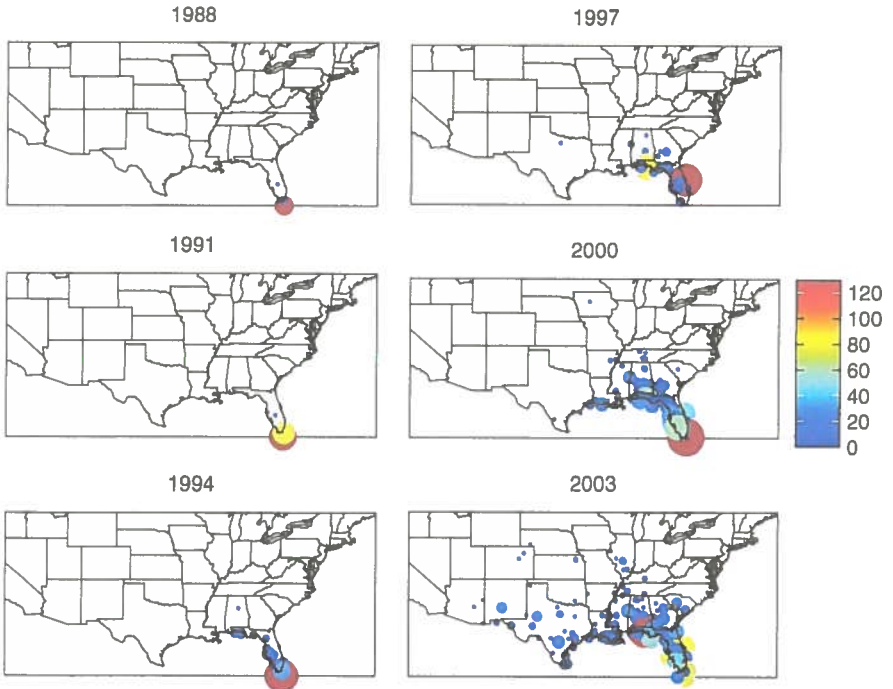


Figure 5.5 Yearly counts (at 3-year intervals) of the Eurasian Collared Dove, obtained from the North American Breeding Bird Survey. The color and radius of the circles are proportional to the counts.

5.1.3 Empirical Covariance/Correlation Functions

It can be quite informative to examine plots of empirical spatio-temporal covariance (or correlation) matrices. For example, assume we have observations $\mathbf{Z}_t \equiv (Z(\mathbf{s}_1; t), \dots, Z(\mathbf{s}_m; t))'$, for $t = 1, \dots, T$. An $m \times m$ *empirical* (averaged over time) *lag- τ spatial covariance matrix* is given by

$$\hat{\mathbf{C}}_Z^{(\tau)} \equiv \frac{1}{T - \tau} \sum_{t=\tau+1}^T (\mathbf{Z}_t - \hat{\boldsymbol{\mu}}_Z)(\mathbf{Z}_{t-\tau} - \hat{\boldsymbol{\mu}}_Z)', \quad \tau = 0, 1, \dots, T - 1, \quad (5.1)$$

where the *empirical spatial mean*, $\hat{\boldsymbol{\mu}}_Z$, is given by

$$\hat{\boldsymbol{\mu}}_Z \equiv \frac{1}{T} \sum_{t=1}^T \mathbf{Z}_t. \quad (5.2)$$

Notice that in (5.1) we choose to divide by $(T - \tau)$ rather than T . If we divide by T , the sequence of empirical lag- τ ($\tau = 0, 1, \dots, T - 1$) covariance

matrices is guaranteed to be nonnegative definite. We use the divisor $(T - \tau)$ here because it more effectively mitigates against bias when a presumed stationary covariance is being estimated. Furthermore, the *empirical lag- τ spatial correlation matrix* is given by

$$\hat{\mathbf{R}}_Z^{(\tau)} \equiv \hat{\mathbf{D}}_Z^{-1/2} \hat{\mathbf{C}}_Z^{(\tau)} \hat{\mathbf{D}}_Z^{-1/2}, \quad (5.3)$$

where $\hat{\mathbf{D}}_Z \equiv \text{diag}(\hat{\mathbf{C}}_Z^{(0)})$ is a diagonal matrix with spatially indexed empirical variances (defined by averaging over time) on the main diagonal.

As an example, consider the empirical lag-0 covariance and correlation matrices for SST anomalies for locations in a domain *along the equator* in the western Pacific Ocean. A useful way to present these matrices is as an image plot. Figure 5.6 shows image plots for the empirical lag-0 covariance (given by (5.1) with $\tau = 0$) and correlation (given by (5.3) with $\tau = 0$) matrices for the SST anomalies. The covariance plot (Figure 5.6a) shows clearly that the variability is greater in the eastern portion of the domain and that there is a fairly strong spatial dependence throughout the domain. The associated correlation plot (Figure 5.6b) shows that there is negative dependence between the extreme western portion of the domain and the eastern portion of the domain. It is also informative to check for spatio-temporal dependence at lags $\tau = 1, 2, \dots$. Since ocean dynamics operate on relatively slow time scales, it is reasonable to expect dependence over quite large time lags. Consider the empirical lag-6 (month) covariance and correlation matrices for the equatorial SST anomalies, shown in Figure 5.7. They demonstrate very strong positive 6-month-lag dependence in the western portion of the domain. Note that, based on the definition of the cross-covariance and cross-correlation functions, these lag matrices are not symmetric, but $\hat{\mathbf{C}}_Z^{(\tau)} = (\hat{\mathbf{C}}_Z^{(-\tau)})'$. A similar relation holds for lag-correlation matrices.

It is also of interest to compute the empirical cross-covariances or cross-correlations between different spatio-temporal variables. For example, consider two data sets, $\mathbf{Z}_t \equiv (Z(\mathbf{s}_1; t), \dots, Z(\mathbf{s}_m; t))'$ and $\mathbf{X}_t \equiv (X(\mathbf{x}_1; t), \dots, X(\mathbf{x}_\ell; t))'$, for $t = 1, \dots, T$, where the locations in $\mathbf{Z}_t$ and $\mathbf{X}_t$ need not coincide, and these vectors need not be of the same dimension. An *empirical lag- τ cross-covariance matrix* is given by

$$\hat{\mathbf{C}}_{\mathbf{Z}, \mathbf{X}}^{(\tau)} \equiv \frac{1}{T - \tau} \sum_{t=\tau+1}^T (\mathbf{Z}_t - \hat{\boldsymbol{\mu}}_Z)(\mathbf{X}_{t-\tau} - \hat{\boldsymbol{\mu}}_X)', \quad (5.4)$$

where recall that the empirical spatial mean, $\hat{\boldsymbol{\mu}}_Z$, is given by (5.2), and likewise

$$\hat{\boldsymbol{\mu}}_X \equiv \frac{1}{T} \sum_{t=1}^T \mathbf{X}_t. \quad (5.5)$$

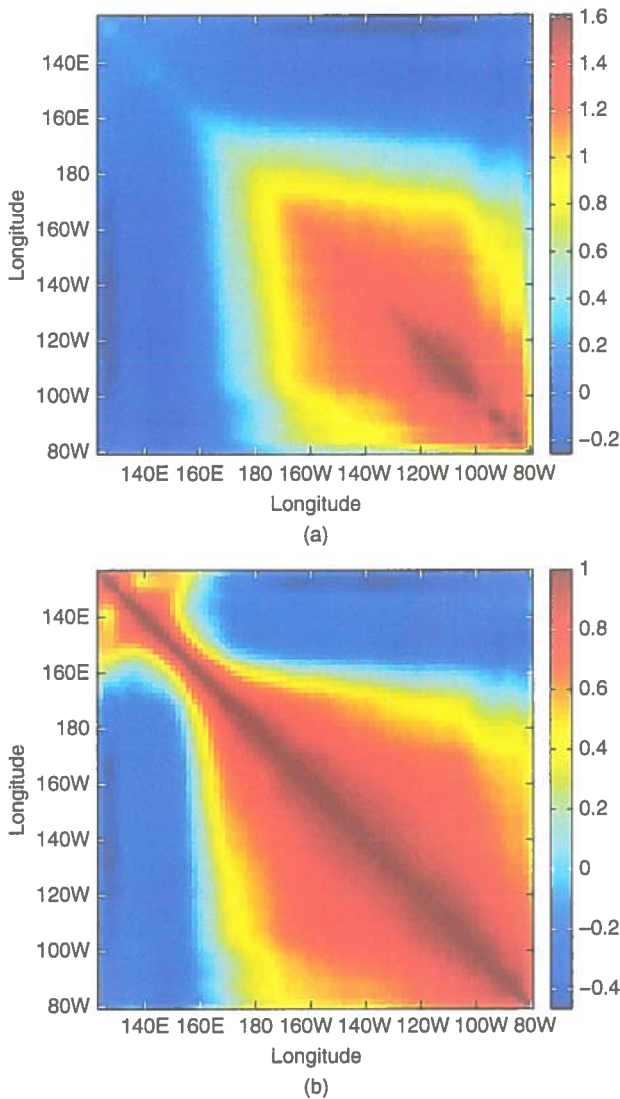


Figure 5.6 Image plots of (a) the empirical lag-0 covariance matrix and (b) the empirical lag-0 correlation matrix for the sea surface temperature anomalies along the equator in the Pacific region.

Note that the empirical lag- τ cross-covariance matrix (5.4) is an $m \times \ell$ matrix, as is the empirical lag- τ spatial cross-correlation matrix given by

$$\hat{\mathbf{R}}_{Z,X}^{(\tau)} \equiv \hat{\mathbf{D}}_Z^{-1/2} \hat{\mathbf{C}}_{Z,X}^{(\tau)} \hat{\mathbf{D}}_X^{-1/2}, \quad (5.6)$$

where $\hat{\mathbf{D}}_Z \equiv \text{diag}(\hat{\mathbf{C}}_Z^{(0)})$ and $\hat{\mathbf{D}}_X \equiv \text{diag}(\hat{\mathbf{C}}_X^{(0)})$.

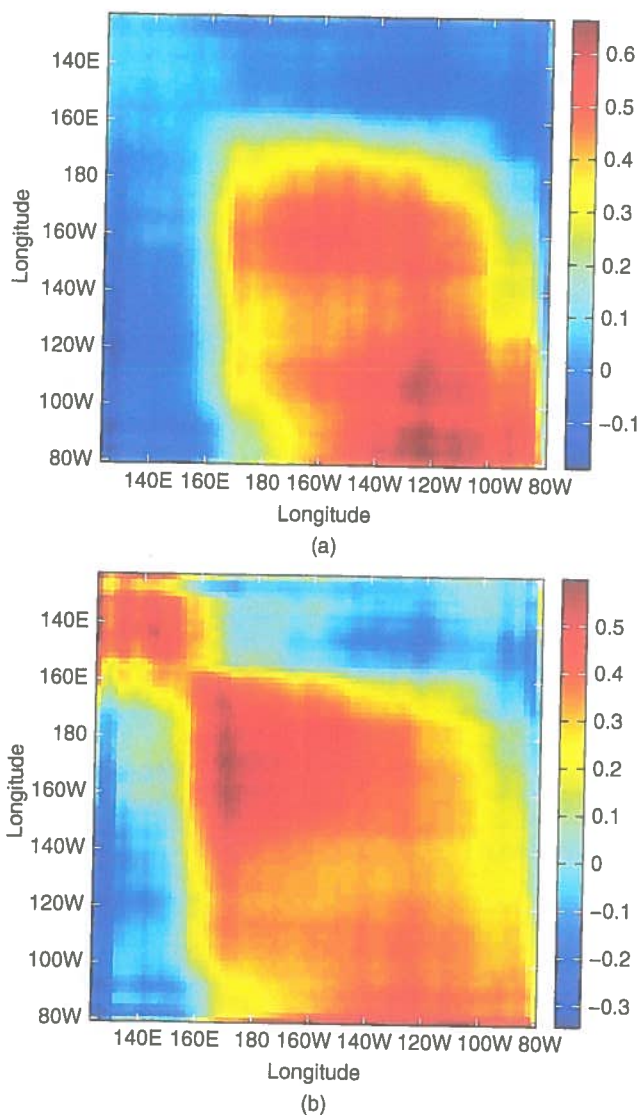


Figure 5.7 Image plot of (a) the empirical lag-6 covariance matrix and (b) the empirical lag-6 correlation matrix for the sea surface temperature anomalies along the equator in the Pacific region.

For example, Figure 5.8 shows the lag-6 cross-correlation for the special case of $\ell = 1$, where the X -variable is the near-surface zonal (i.e., east–west) wind component at (155° E, 5° N) and the Z -variable is the monthly SST anomaly for each of the $m = 2,261$ $2^\circ \times 2^\circ$ pixels in the tropical Pacific region. The time index t increases in one-month increments from January 1970 through December 2000. In this case, since we are computing the lag-6 cross-correlation

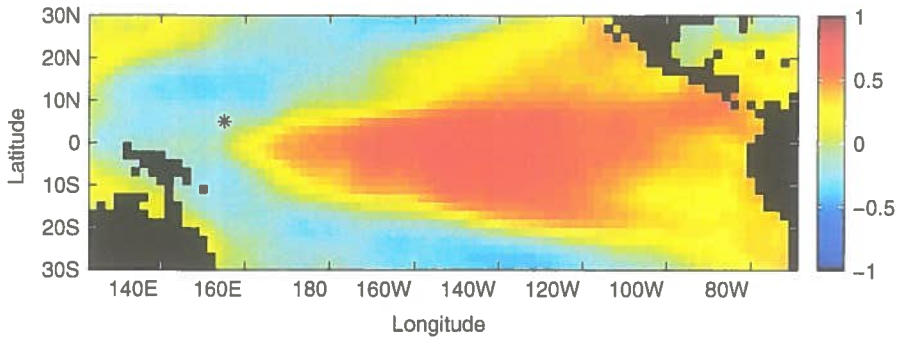


Figure 5.8 Image plot of the lag-6 cross-correlation between the near-surface zonal wind component at 155° E, 5° N (shown with an asterisk) and the monthly sea surface temperature anomalies from January 1970 to December 2000. In this case, the time series of the near-surface zonal wind component lags the sea surface temperature anomalies by 6 months.

for a spatio-temporal field relative to a time series for one spatial location (i.e., $\ell = 1$), we can plot the results as a spatial map. Note that the plot shows a very high correlation between the zonal wind component at this location and the Pacific SST anomalies at spatially coherent locations. In particular, we note a clear El Niño-type pattern in the lag-correlations, suggesting that the zonal wind component at $(155^\circ$ E, 5° N) could be a useful 6-month-ahead predictor of a future El Niño event (e.g., see Berliner, Wikle, and Cressie, 2000). This is incorporated into a BHM presented in Section 9.1.

As we shall see in Section 6.2, the characterization of the joint spatio-temporal covariance structure of a spatio-temporal process is important for optimal prediction. Thus, it is important to examine the empirical covariance function at various space and time lags. Making the (implicit) assumption that the first moment depends on space but not on time, and the second moment depends only on spatial and temporal *lag differences*, then the estimated spatio-temporal covariance at spatial lag $\mathbf{h}$ and time lag τ is given by

$$\hat{C}_Z(\mathbf{h}; \tau) \equiv \frac{1}{|N_s(\mathbf{h})|} \frac{1}{|N_t(\tau)|} \sum_{\mathbf{s}_i, \mathbf{s}_j \in N_s(\mathbf{h})} \sum_{t, r \in N_t(\tau)} (Z(\mathbf{s}_i; t) - \hat{\mu}_Z(\mathbf{s}_i))(Z(\mathbf{s}_j; r) - \hat{\mu}_Z(\mathbf{s}_j)), \quad (5.7)$$

where recall from (5.2) that $\hat{\mu}_Z(\mathbf{s}_i) \equiv (1/T) \sum_{t=1}^T Z(\mathbf{s}_i; t)$, $N_s(\mathbf{h})$ refers to the pairs of spatial locations with spatial lag within some tolerance of $\mathbf{h}$, $N_t(\tau)$ refers to the pairs of time points with time lag within some tolerance of τ , and $|N(\cdot)|$ refers to the cardinality (i.e., number of elements) of the set $N(\cdot)$. Note that, as with purely spatial covariogram and variogram estimation (Section 4.1.1), potentially anisotropic spatio-temporal dependence can be investigated by allowing spatial lags $\mathbf{h}$ to have various directions. From (5.7), we can construct the lag- τ empirical covariance matrices $\hat{\mathbf{C}}^{(\tau)}$; $\tau = 0, 1, 2, \dots$

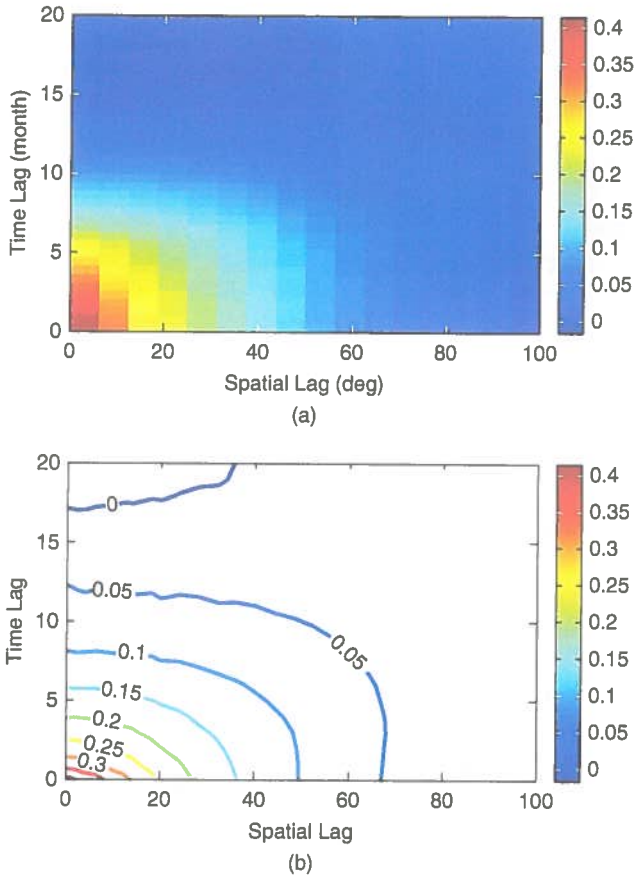


Figure 5.9 Empirical spatio-temporal covariance plots for the sea surface temperature anomalies in the tropical Pacific region (January 1970 to December 2002). The y axis shows temporal lag in months and the x axis shows spatial lag in degrees. (a) An image plot. (b) A contour plot.

Figure 5.9 shows the empirical spatio-temporal covariance function for temporal lag τ in months and the modulus of the spatial lag $h \equiv \|\mathbf{h}\|$ in degrees, for the tropical Pacific SST anomaly dataset from January 1970 through December 2002. Of course, we are free to visualize this information in different ways with our favorite types of plots. For example, we show both an image plot and a contour plot for illustration. See Sections 6.1 and 6.2 for more discussion about how spatio-temporal covariance plots can be used to characterize spatio-temporal dependence.

5.1.4 Spatio-Temporal LISAs

Local Indicators of Spatial Association (LISAs) decompose global statistics into components where each component is associated with a spatial coordinate

(Anselin, 1995) or, in our case, with a spatio-temporal coordinate. The important property of a LISA is that an “average” (in the very general sense of the word) of the local components yields the global component. In a purely spatial setting, a LISA applied to an empirical covariance function can show unusual components at various locations, thereby allowing a visualization of spatial outliers or departures from stationarity. Cressie and Collins (2001) study LISAs for empirical pair correlation functions of spatial point patterns.

Examples of LISAs: In a spatio-temporal setting, consider the yearly empirical covariance matrix $\hat{C}_Z^{(0)}$ for SST anomalies, which is a 12-monthly average of cross-products; each of these local (in time) averages is taken over January through December in any given year from 1970 through 2002. A plot of the leading eigenvalues of these *spatial* covariance matrices versus year would investigate a type of local (yearly) temporal variability that is ignored in a typical principal components analysis. As we saw in Section 4.1.1 and will see in much more detail in Section 5.3, a principal components analysis (i.e., an eigenvector/eigenvalue decomposition) of a spatial covariance matrix is often termed an EOF analysis in the geophysics literature. In this case, the eigenvectors correspond to patterns of maximal variability, with the largest eigenvalue corresponding to the “dominant pattern” of variability over the time period used to calculate the spatial covariance matrix. We now visualize the leading eigenvalue and its associated eigenvector as a function of year, from 1970 to 2002. Each yearly value and map could be considered as a LISA because they approximate the empirical spatial covariance matrix from that year, and the overall empirical spatial covariance matrix is linearly related to the yearly empirical covariance matrices.

A LISA plot of the leading eigenvalue for the SST data for consecutive 12-month periods is shown in Figure 5.10. One can see clearly from this plot that the leading EOF varies greatly across the 33-year period. In addition, the years in which the leading eigenvalue is greatest typically corresponds to years in which the El Niño and La Niña phenomena are most active. To illustrate this, Figure 5.11 shows the corresponding leading eigenvector (EOF) maps for 1981 and 1982. Note that the 1981 EOF does not show the clear ENSO structure that is apparent in 1982. The abrupt transition into an ENSO state is evidence of the likely lack of temporal stationarity in these data.

As another example, consider the *temporal* empirical covariance function for the *monthly* SST anomaly data from January 1970 to December 2002, for each $2^\circ \times 2^\circ$ pixel. Figure 5.12 shows a spatial plot of the temporal empirical autocovariances at lag $\tau = 1$ (month). Again, there is noticeably higher temporal autocovariance in the ENSO region of the eastern Pacific, suggesting a lack of spatial stationarity in the lag-1 temporal autocovariances. These visualizations show that a geostatistical analysis of SST that assumes stationarity in space and/or time (Section 6.1.2) would be inappropriate.

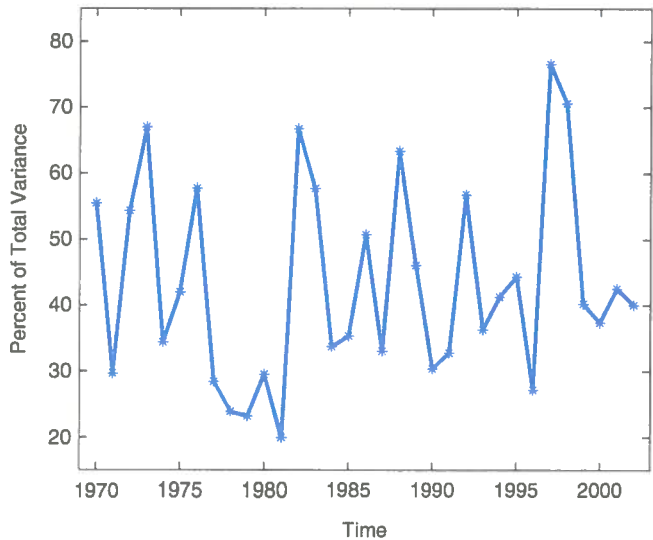


Figure 5.10 LISA plot showing the eigenvalue of the leading EOF for the monthly sea surface temperature anomaly data for consecutive 12-month periods starting in January 1970. For presentation purposes, the leading EOF eigenvalue is normalized by the total variance to show the percent variance accounted for by the leading EOF for each year.

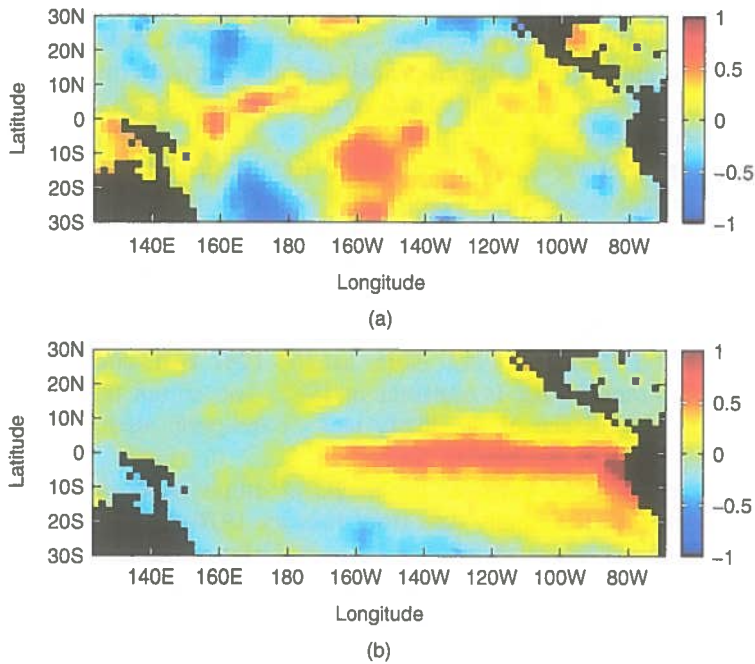


Figure 5.11 The spatial map of the leading EOF of the monthly sea surface temperature anomaly data associated with the LISA analysis in Figure 5.10. (a) 1981. (b) 1982.

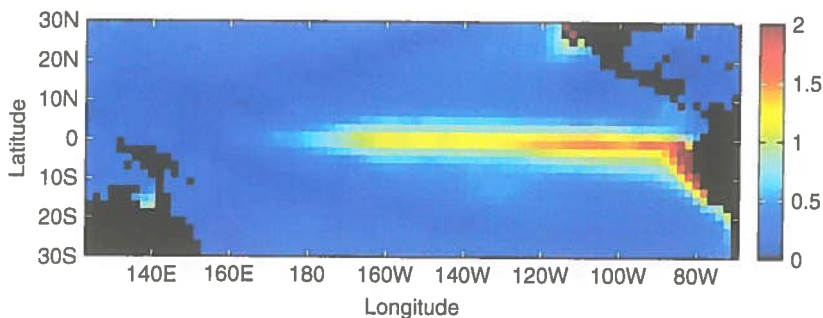


Figure 5.12 LISA plot showing the lag-1-month temporal autocovariance at each spatial location for the monthly sea surface temperature anomaly data.

5.1.5 Spatio-Temporal Parallel Coordinates Plot

The *Parallel Coordinates* (PC) plot is a way of visualizing multivariate data (Inselberg, 1985; Wegman, 1990). Visually, the vertical axes are placed in parallel in some predetermined order. Each component of a given multivariate observation is plotted on its respective axis. Then a piecewise straight line is drawn between corresponding values on each axis. For many multivariate observations, the PC plot shows an array of piecewise straight lines that may be difficult to interpret. An enhancement is to compare multivariate dependence between different classes by coloring their respective arrays with different colors.

Now consider spatio-temporal multivariate data $\{Z_t(s_i) : s_i \in D_s, t = 1, 2, \dots\}$. The obvious use of PC plots for such data is to treat the observations as multivariate (spatial) vectors in time. For example, say there are observations at $i = 1, \dots, m$ spatial locations for each time t . Recall that we denoted these as vectors Z_t . In this case, varying the index t generates a “sample,” $Z_1, \dots, Z_T$, of m -variate vectors. Hence, each vertical axis of the PC plot represents a spatial location, and the corresponding T values associated with a location are plotted on the vertical axis. Consequently, this spatio-temporal PC plot shows T lines, each one representing $m - 1$ connected line segments. A potential difficulty with such a representation is that with large m and T , it may be difficult to extract much information from these plots due to the “visual clutter” inherent with greater line density (e.g., Edsall, 2003). One approach to reduce this clutter is to collapse the m spatial locations into fewer components, and a natural way to do this in the spatio-temporal context is to project the data onto the first p_α EOFs (e.g., see Section 5.3.2). The associated principal component (p_α -variate) time series associated with these projections becomes the new dataset. In this case, the data are represented in terms of p_α -variate vectors, which suggests that a vertical axis now represents a particular EOF rather than a location in physical space, and the ordering of the axes is naturally from the first EOF to the p_α th.

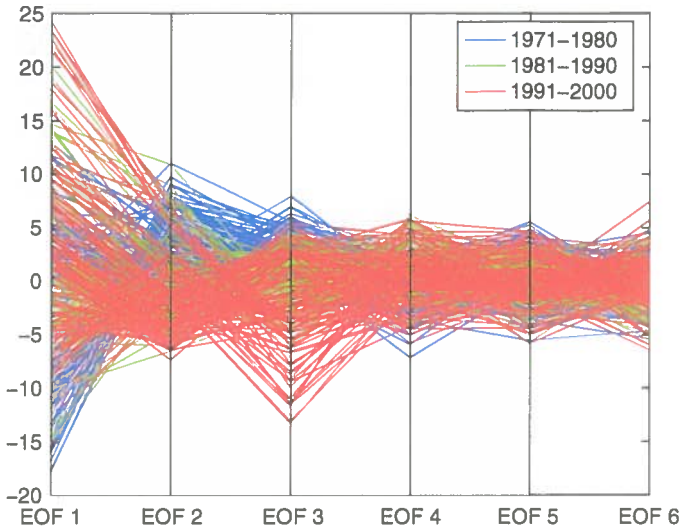


Figure 5.13 Parallel coordinates (PC) plot for the monthly sea surface temperature anomalies in the tropical Pacific region. The data were projected onto the first six EOFs and the associated principal components represent the x axis coordinates. Each connected line represents the values of these EOF variables for a given month (over the 30-year period from January 1971 to December 2000). Lines associated with the decades are indicated by different colors (blue: 1971–1980; green: 1981–1990; red: 1991–2000).

Example: Figure 5.13 shows a PC plot for the monthly SST anomaly dataset for the 30-year period from January 1971 to December 2000. Specifically, this plot shows the PC plot based on the projection of the data onto the first six EOFs, and there are $T = 12 \times 30 = 360$ piecewise straight lines superimposed. The use of color to differentiate the decades (1971–1980: blue; 1981–1990: green; 1991–2000: red) shows that the loadings on the EOFs are quite different between the first (blue) and last (red) ten-year periods, especially for the first four EOFs. This matches with the example in Section 5.3.4, which illustrates that the first four EOFs are associated with the El Niño/La Niña phenomena, and these are known to vary substantially across decades.

There are other ways that one could use PC plots to analyze spatio-temporal data. Of course, the approach described above could be used, but with T -dimensional vectors $\mathbf{Z}(\mathbf{s}_i)$; $i = 1, \dots, m$, so that the vertical axes now represent successive time points. This is a useful way to examine multivariate time series in general but suffers from the same “visual clutter” problem described when m is large. However, projecting onto the p_α leading (spatial) EOFs would reduce the number of lines from m to p_α (e.g., $m = 2,261$ and $p_\alpha = 6$).

For spatio-temporal data, we have had success with a linked PC plot, where the PC plot is based on T -variate vectors $\{\mathbf{Z}(\mathbf{s}_i) : i = 1, \dots, m\}$, and it is linked dynamically to a map showing spatial coordinates $\{\mathbf{s}_i : i = 1, \dots, m\}$.

By highlighting (brushing) either in the PC-plot view or in the spatial view, there is linked highlighting in the other view; see Ahlqvist et al. (2010).

5.2 SPECTRAL ANALYSIS

Building on the univariate and bivariate spectral-analysis methods discussed in Sections 3.5.3 and 3.5.4, one can use spectral analysis to gain insight into spatio-temporal data as well. This can be accomplished by several exploratory methods. Here, we focus on the use of cross-spectral analysis between several data locations separated in space (Section 5.2.1), as well as so-called *spatio-temporal spectral analysis* (Section 5.2.2).

5.2.1 Cross-Spectral Analysis

Careful application of the basic cross-spectral-analysis methods outlined in Section 3.5.4 can be used to explore spatio-temporal data. In particular, such methods can be used to discover properties of spatio-temporal features such as propagating waves. Wikle, Madden, and Chen (1997) provide a cross-spectral analysis of upper atmospheric winds in a study of tropical waves in the atmosphere. Of particular interest to these authors was a type of equatorial wave in the lower stratosphere known as a *mixed Rossby-gravity wave* (MRGW). These waves have a westward phase propagation and a period of about 3–5 days.

Recall that in Section 3.5.4 we considered the cross-spectral analysis between the u - and v -components of the wind in the lower stratosphere for a station (on Chuuk Island) in the tropical Pacific (see Figure 3.22). Recall also that the u -component is the east/west (or zonal) component and that the v -component is the north/south (or meridional) component. Figure 3.22 shows a squared coherence peak in the frequency range corresponding to the MRGW and, more importantly, showed that the v -component of the wind leads the u -component by a quarter of a cycle, in remarkable agreement with equatorial wave theory (e.g., Matsuno, 1966). However, the cross-spectral analysis at a single station does not provide information from which to infer the movement of these waves. For that, one must consider time series from multiple spatial locations.

To illustrate our point, we use once-daily upper air weather balloon (i.e., rawinsonde) observations during the period, 1 January 1962–31 December 2007 (a different dataset than the one used in Wikle, Madden, and Chen, 1997). We focus on the observations from the 70-hPa level of constant pressure (in the lower stratosphere, approximately 18.5 km above the surface). We considered the daily data at the four tropical Pacific stations (Koror, Chuuk—formerly Truk, Ponape, and Majuro) shown in Figure 5.14; all of these stations are in a fairly narrow latitude band.

To study the average propagation properties of wavelike disturbances in this dataset, consider the cross-spectral magnitude squared coherence (MSC)

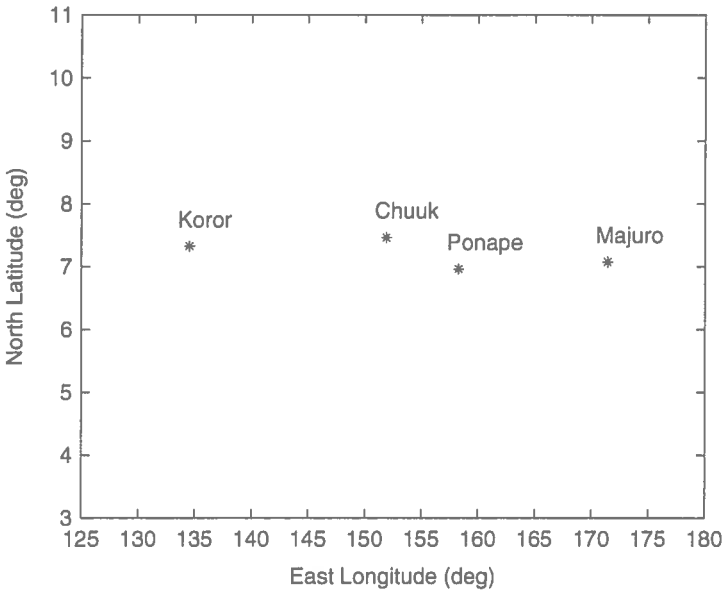


Figure 5.14 Locations of tropical stations for the Western Pacific equatorial wave cross-spectral analysis. The y axis and x axis show $^{\circ}$ N latitude and $^{\circ}$ E longitude, respectively.

and the phase spectrum (see Section 3.5.4 for their definitions) between the v -components at each of the six possible combinations of the four stations (Koror, Chuuk, Ponape, and Majuro). A portion of the MSC (top panel) and phase spectrum (bottom panel) plots are shown in Figure 5.15.

The figure shows that the MSC between the various pairs of stations are relatively large throughout this portion of the frequency domain, although it is generally smaller as station separation increases. The associated phase spectra are quite different, but suggest propagation. For instance, at 0.25 cycles per day (cpd), the Chuuk–Majuro phase is negative, implying that the Majuro v -component leads that at Chuuk. Similarly, the Chuuk–Koror phase at the same frequency is positive, implying that the Chuuk v -component leads that at Koror. Since Koror is west of Chuuk, and Chuuk is west of Majuro, the average phase propagation must be toward the west. Furthermore, the magnitudes of these phase differences are indicative of the horizontal scale of the wave. Wikle, Madden, and Chen (1997) show that one can use this information to perform more formal inference concerning wave properties.

5.2.2 Spatio-Temporal Spectral Analysis

Spectral analysis of time series is discussed in Section 3.5. In the spatial setting, two-dimensional spectral analysis has been used to explore both the structure and patterns of spatial datasets (e.g., Renshaw and Ford, 1983). It is natural to

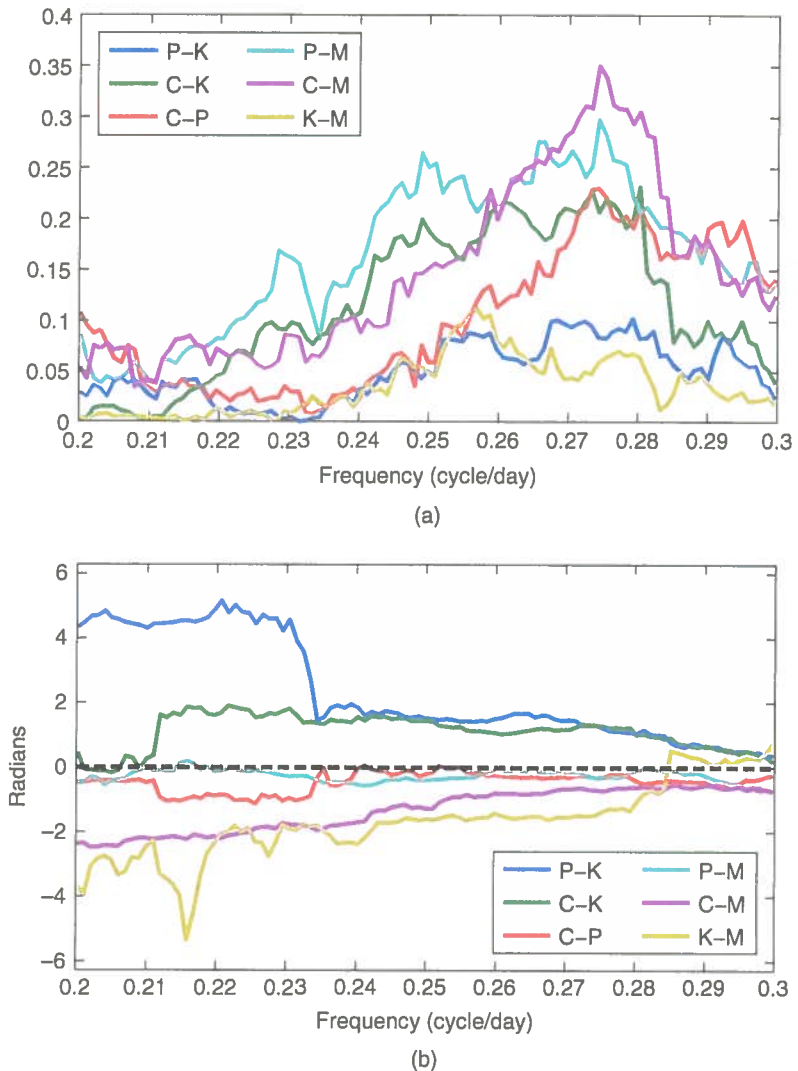


Figure 5.15 Cross-spectral analysis between the daily 70-hPa v -wind component at 0:00 UTC (midnight Greenwich Mean Time) at pairs of tropical Pacific stations for the period from 1 January 1962 through 31 December 2007. Results were obtained using a smoothed periodogram with smoothing bandwidth of 0.04 cycles per day. (a) The magnitude squared coherence (MSC) (see Section 3.5.4) for the pairs of stations. (b) The corresponding phase spectrum (in radians). Note that only a portion of the frequency domain is shown. Station labels are given by the first letter of the station name as shown in Figure 5.14. For example, “P-K” represents the MSC or phase spectrum between v -wind components at Ponape and Koror.

consider a hybrid of these methodologies for data that are referenced in both space and time. As discussed in Section 3.5 for temporal processes, in Section 4.1.1 for spatial processes, and in Section 6.1.5 for spatio-temporal processes, there is direct correspondence between the covariance function and the spectrum. Thus, it can be informative to explore the spatio-temporal spectrum of a dataset in order to gain some appreciation for its dependence structure in space and time. For example, such analyses are useful in climatology and meteorology due to the prominence of wave-like processes in their governing dynamics. In particular, in order to make inference about relationships between spatial and temporal scales of variability, such data are often analyzed by a variant of two-dimensional spectral analysis in which time and one spatial dimension are considered (e.g., Hayashi, 1971, 1982a, 1982b; Speth and Madden, 1983). This type of spectral analysis is not common in the Statistics literature, and thus we outline the basics of the approach here.

In Atmospheric Science, this methodology is typically applied to data in which the spatial axis is longitude and “positive (negative)” corresponds to eastward (westward) propagation. Assume that a given spatio-temporal process is cyclic in one-dimensional space $s \in D_s$ (e.g., s corresponds to longitude and s is positive to the east) and limited to discrete time $t \in D_t = \{1, \dots, T\}$. Denote the data by $\{Z_t(s) : s \in D_s, t \in D_t\}$, and expand it in Fourier harmonics in the spatial dimension with *wavenumber* k :

$$Z_t(s) = m_t + \sum_{k=1}^{\infty} \{a_{t,k} \cos(ks) + b_{t,k} \sin(ks)\}, \quad (5.8)$$

where m_t represents the spatial average for each time t , and $a_{t,k}$ and $b_{t,k}$ are the time-varying spatial cosine and sine Fourier transform coefficients, respectively. The spatio-temporal power spectrum, P , of positive traveling (eastward if s refers to longitude) and negative traveling (westward if s refers to longitude) waves can be computed according to

$$P(k, \pm \omega) = \frac{1}{4} [f_a(k, \omega) + f_b(k, \omega) \pm 2q_{a,b}(k, \omega)], \quad (5.9)$$

where the $+$ and $-$ refer to positive and negative traveling waves, respectively, ω is the time frequency, $f_a(k, \omega)$ and $f_b(k, \omega)$ are the temporal power spectra for $\{a_{k,t} : t = 1, \dots, T\}$ and $\{b_{k,t} : t = 1, \dots, T\}$, respectively, and $q_{a,b}(k, \omega)$ is the quadrature spectrum of the time series $\{a_{t,k}\}$ and $\{b_{t,k}\}$ (see Section 3.5.4 for the definition of the quadrature spectrum). This formula is a generalization of Deland's (1964, 1972) observation that since the cosine and sine coefficients of traveling waves are 90° out of phase, the quadrature spectrum indicates the squared amplitude of traveling waves, and its sign indicates the direction of phase propagation. The spatio-temporal power spectrum of the standing oscillation, ST , for the k th spatial wavenumber and time-frequency ω , can

then be expressed as

$$ST(k, |\omega|) = \left\{ \frac{1}{4} [f_a(k, \omega) - f_b(k, \omega)]^2 + c_{a,b}^2(k, \omega) \right\}^{1/2}, \quad (5.10)$$

where $c_{a,b}(k, \omega)$ is the co-spectrum (see Section 3.5.4 for more details) of the time series $\{a_{t,k}\}$ and $\{b_{t,k}\}$. It is then possible to isolate the corresponding “pure” eastward and westward propagating components, PU , as

$$PU(k, \pm \omega) = P(k, \pm \omega) - \frac{1}{2} ST(k, |\omega|). \quad (5.11)$$

As outlined by Hayashi (1982a), this method of partitioning space-time power spectra into standing and traveling parts is based on the following definitions and assumptions:

- Standing waves are defined as consisting of eastward and westward moving components that are of equal amplitude and have high coherence (see Section 3.5.4 for the definition of the coherence measure).
- Traveling waves are defined as consisting of eastward and westward moving components that have low coherence with each other.
- Standing and traveling waves are of a different origin and have low coherence with each other.

The validity and applicability of these assumptions are often called into question; for a discussion, see von Storch and Zwiers (1999, pp. 242–244). In addition to considering the validity of these assumptions, in practice one must consider all of the usual concerns of one-dimensional spectral analysis (e.g., stationarity, end effects, aliasing, statistical consistency, window length, etc., as discussed in Section 3.5).

Empirical Spatio-Temporal Spectrum

Calculation of the spatio-temporal spectrum proceeds using the basic approaches for classical univariate and bivariate spectral analysis described in Section 3.5. Specifically, consider spatio-temporal data from a lattice in one-dimensional space and discrete, equally spaced time points, $\{Z_t(\mathbf{s}_i): i = 1, \dots, m, t = 1, \dots, T\}$, where for simplicity of presentation, we assume that m and T are both even numbers. One algorithm proceeds as follows:

1. Take the *spatial* discrete Fourier transform (DFT) of the elements in the m -dimensional vector, $\mathbf{Z}_t \equiv (Z_t(\mathbf{s}_1), \dots, Z_t(\mathbf{s}_m))'$, for each time, t . Denote the cosine and sine coefficients from these DFTs as $\hat{a}_{t,k}$ and $\hat{b}_{t,k}$, respectively, where k refers to the spatial wave number; $k = 0, 1, \dots, m/2$.

2. Calculate the power spectra and cross-spectra for the time series from the calculations given in step 1.
 - (a) Calculate the temporal power spectra for each of the sine and cosine wavenumber time series from step 1 using the smoothed periodogram approach (e.g., see Section 3.5.3). Denote these as $\hat{f}_a(k, \omega_j)$ and $\hat{f}_b(k, \omega_j)$, for wavenumber $k = 0, \dots, m/2$ and temporal frequency $\omega_j = 2\pi j/T$; $j = 1, \dots, T$.
 - (b) Calculate the temporal cross-spectrum between the sine and cosine wavenumber time series using the smoothed cross-periodogram approach (e.g., see Section 3.5.4). Denote this by $\hat{f}_{a,b}(k, \omega_j)$, where the real and imaginary components give the associated co-spectra and quadrature spectra, respectively: $\hat{f}_{a,b}(k, \omega_j) = \hat{c}_{a,b}(k, \omega) - i\hat{q}_{a,b}(k, \omega)$, where $i \equiv \sqrt{-1}$.
3. Calculate the spatio-temporal power-spectrum components from the calculations given in step 2.
 - (a) Calculate the spatio-temporal power-spectrum components (5.9) for eastward (positive) and westward (negative) traveling waves, respectively:

$$\hat{P}(k, +\omega) = \frac{1}{4}[\hat{f}_a(k, \omega) + \hat{f}_b(k, \omega) + 2\hat{q}_{a,b}(k, \omega)], \quad (5.12)$$

$$\hat{P}(k, -\omega) = \frac{1}{4}[\hat{f}_a(k, \omega) + \hat{f}_b(k, \omega) - 2\hat{q}_{a,b}(k, \omega)]. \quad (5.13)$$

- (b) Calculate the spatio-temporal power-spectrum of the standing oscillation component (5.10):

$$\hat{S}T(k, |\omega|) = \left\{ \frac{1}{4}[\hat{f}_a(k, \omega) - \hat{f}_b(k, \omega)]^2 + \hat{c}_{a,b}^2(k, \omega) \right\}^{1/2}. \quad (5.14)$$

- (c) Calculate the spatio-temporal power-spectrum components (5.11) for the “pure” eastward and westward propagating components, respectively:

$$\hat{P}U(k, +\omega) = \hat{P}(k, +\omega) - \frac{1}{2}\hat{S}T(k, |\omega|), \quad (5.15)$$

$$\hat{P}U(k, -\omega) = \hat{P}(k, -\omega) - \frac{1}{2}\hat{S}T(k, |\omega|). \quad (5.16)$$

Example: Once again, consider the tropical Pacific SST data discussed in Section 5.1. In particular, consider the monthly SST anomalies averaged over the region from 1° South to 1° North for the equatorial Pacific region (see Figure 5.1). We now consider the spatio-temporal spectral analysis of these data over the 33-year period from January 1970 to December 2002; the empirical spectral density for these data are shown by the contour plots in Figure 5.16. The top panel in this plot shows the total power spectrum, $\hat{P}(k, \pm\omega)$, for

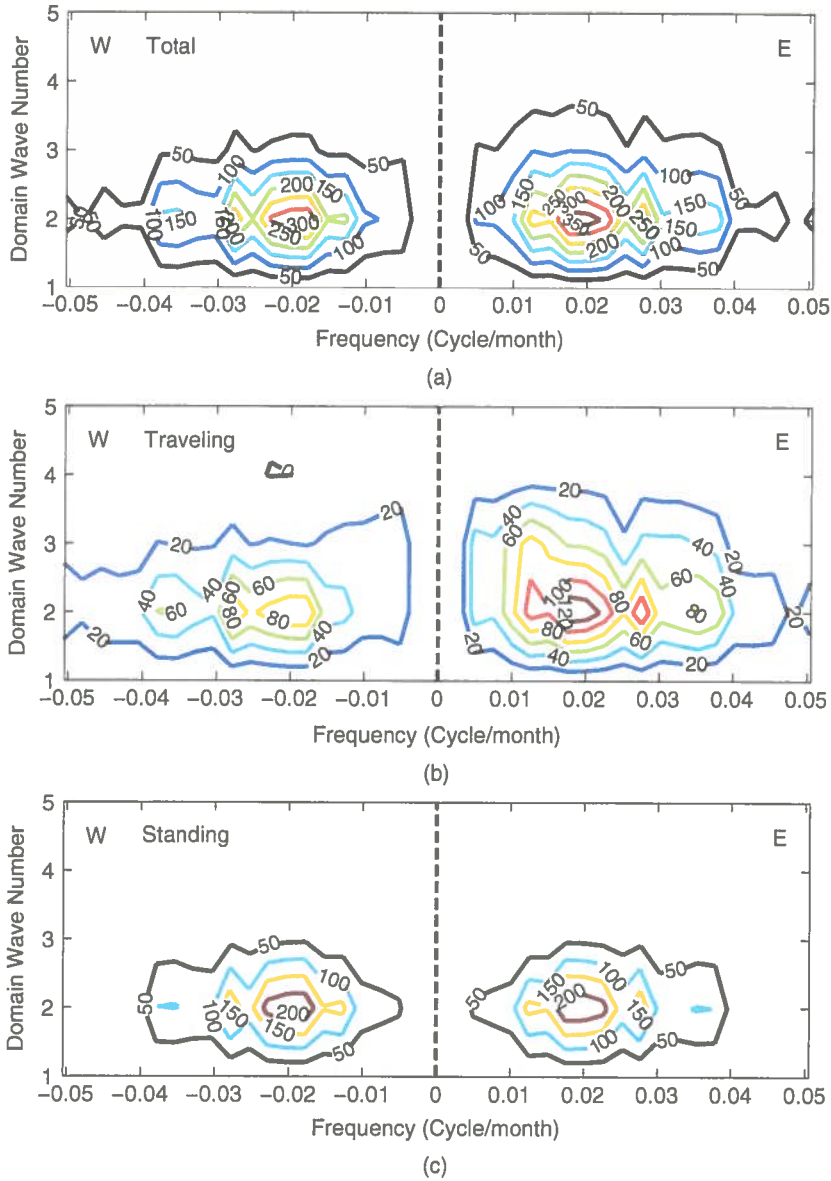


Figure 5.16 Spatio-temporal spectral analysis for SST anomalies at the equator (averaged from 1° N to 1° S, across the Pacific region). (a) Total power spectral density ($\hat{P}$). (b) Pure traveling power spectral density ($\hat{P}U$). (c) Standing power spectral density ($\hat{S}T$).

a portion of the temporal frequency domain (x axis) and the wave-number domain (y axis). The left portion of the plot shows the results for negative (westward propagating) frequencies and the right portion shows the results for the positive (eastward propagating) frequencies. Note that the power associated with the westward- and eastward-propagating waves is not symmetric; specifically, there is more power associated with westward-propagating waves.

As discussed above, the power-spectral density can be decomposed into pure travelling waves and pure standing waves. These components, $\hat{P}U(k, \pm \omega)$ and $\hat{S}T(k, \omega)$, are shown in the middle and bottom panels of Figure 5.16, respectively. Clearly, most of the power is associated with the traveling waves; within these, the most power is associated with oscillations at frequencies of about ± 0.026 cycles per month (i.e., a period of 38.5 months or 3.2 years), which is essentially the average frequency of the ENSO phenomena. In addition, we see that the power is centered around the (domain) wavenumber of about 2 to 3. That is, we would see between 2 to 3 waves (in space) over the equatorial Pacific region, since spatial “wavenumber” is defined relative to the longitudinal spatial extent of the domain D_s .

5.3 EMPIRICAL ORTHOGONAL FUNCTION (EOF) ANALYSIS

Empirical orthogonal functions (EOFs) are the geophysicist’s terminology for the eigenvectors in the classical eigenvalue/eigenvector decomposition of a covariance (or a correlation) matrix. In its discrete formulation, EOF analysis is simply Principal Component Analysis or PCA (e.g., Hotelling, 1933; Jolliffe, 2002), while in its spatially continuous formulation, it is a Karhunen–Loève (K–L) expansion (e.g., Loève, 1963) as described in Section 4.1.4. Depending on the application, EOFs are usually used (1) in a diagnostic mode to find principal (in terms of explanation of variance) spatial structures, along with the corresponding time variation of those structures, and (2) to reduce the dimension (spatially or temporally) in large spatio-temporal datasets, while simultaneously reducing noise. An extensive review of EOFs, their theory, and their application to meteorology and oceanography is contained in Preisendorfer (1988). In addition, von Storch and Zwiers (1999), Jolliffe (2002), and Wilks (2006) give quite complete overviews of EOF analysis.

5.3.1 Spatially Continuous Formulation

We first consider data $\{Z_t(s) : s \in D_s, t = 1, 2, \dots\}$ from a continuous spatial process measured at discrete time intervals. Our goal is to find an optimal and separable orthogonal decomposition of this process; that is, we wish to write

$$Z_t(s) = \sum_{k=1}^{\infty} \alpha_t(k) \phi_k(s), \quad (5.17)$$

such that $\text{var}(\alpha_t(1)) > \text{var}(\alpha_t(2)) > \dots$, and $\text{cov}(\alpha_t(i), \alpha_t(k)) = 0$, for all $i \neq k$. As described in Section 4.1.4, a well known solution to this problem is obtained through a Karhunen–Loève (K–L) expansion (e.g., see Papoulis, 1965, p. 457–461). Suppose that $E(Z_t(\mathbf{s})) = 0$, and define the spatial covariance function as $E(Z_t(\mathbf{s})Z_t(\mathbf{r})) \equiv C_Z^{(0)}(\mathbf{s}, \mathbf{r})$, which need not be stationary in space but is assumed invariant in time. Then the K–L expansion allows the covariance function to be decomposed as follows:

$$C_Z^{(0)}(\mathbf{s}, \mathbf{r}) = \sum_{k=1}^{\infty} \lambda_k \phi_k(\mathbf{s}) \phi_k(\mathbf{r}), \quad (5.18)$$

where $\{\phi_k(\cdot) : k = 1, 2, \dots\}$ are the eigenfunctions and $\{\lambda_k : k = 1, 2, \dots\}$ are the associated eigenvalues of the Fredholm integral equation,

$$\int_{D_s} C_Z^{(0)}(\mathbf{s}, \mathbf{r}) \phi_k(\mathbf{s}) d\mathbf{s} = \lambda_k \phi_k(\mathbf{r}), \quad (5.19)$$

where

$$\int_{D_s} \phi_k(\mathbf{s}) \phi_l(\mathbf{s}) d\mathbf{s} = \begin{cases} 1 & \text{for } k = l, \\ 0 & \text{otherwise.} \end{cases} \quad (5.20)$$

Assuming completeness of the eigenfunctions (see Section 3.5.1), we can then expand $Z_t(\mathbf{s})$ according to

$$Z_t(\mathbf{s}) = \sum_{k=1}^{\infty} \alpha_t(k) \phi_k(\mathbf{s}), \quad (5.21)$$

where we call $\{\phi_k(\mathbf{s}) : \mathbf{s} \in D_s\}$ the k -th EOF, and we often refer to the associated time series $\{\alpha_t(k) : t = 1, 2, \dots\}$ as the k th principal-component time series, or k th “amplitude” time series. This time series is stochastic and is derived from the projection of the process $\{Z_t(\mathbf{s})\}$ onto the EOF basis; specifically,

$$\alpha_t(k) = \int_{D_s} Z_t(\mathbf{s}) \phi_k(\mathbf{s}) d\mathbf{s}, \quad t = 1, 2, \dots \quad (5.22)$$

It is easy to verify that each of these time series is uncorrelated, with variances equal to the corresponding eigenvalues; that is, $E(\alpha_t(i)\alpha_t(k)) = \lambda_k \delta_{ik}$, where δ_{ik} is the Kronecker delta function (equal to one when $i = k$, and equal to zero otherwise).

If we truncate the expansion (5.17) at p_α , yielding

$$Z_{p_\alpha, t}(\mathbf{s}) \equiv \sum_{k=1}^{p_\alpha} \alpha_t(k) \phi_k(\mathbf{s}), \quad (5.23)$$

then it can be shown (e.g., Freiburger and Grenander, 1965) that the finite EOF decomposition minimizes the variance of the truncation error, $E(Z_t(\mathbf{s}) - Z_{p\alpha,t}(\mathbf{s}))^2$, and is thus optimal in this regard when compared to all other bases.

Since data are always discrete, in practice we must solve the Fredholm integral equation (5.19) numerically to obtain the EOF basis functions. Cohen and Jones (1969) and Buell (1972, 1975) give numerical-quadrature solutions to this problem. The numerical-quadrature approaches for discretizing the Fredholm integral equation give estimates for the eigenfunctions and eigenvalues that are weighted according to the spatial distribution of the data locations, but only for the eigenfunctions at the locations $\{\mathbf{s}_1, \dots, \mathbf{s}_n\}$ for which there are data. Obled and Creutin (1986) have presented an elegant spline-based method for discretizing the K–L integral equation and for interpolating the eigenfunctions to locations where data are not available.

5.3.2 Spatially Discrete Formulation

Although the continuous K–L representation of EOFs is often the most realistic from the point of view of modeling a geophysical process, it is rarely considered in applications. This is due simply to the discrete nature of where and when data are observed, and to the difficulty of solving the K–L integral equation. In cases where there are repeat observations (e.g., over time) or some additional information (e.g., deterministic model output or “historical” observations), one can calculate an empirical covariance matrix and perform a PCA analysis. For example, consider a spatially discrete formulation of EOF analysis by using the PCA formulation as given in standard multivariate statistics books (e.g., Johnson and Wichern, 1992), modified according to the spatio-temporal notation we have introduced.

In that case, let $\mathbf{Z}_t \equiv (Z_t(\mathbf{s}_1), \dots, Z_t(\mathbf{s}_m))'$ and define the k th discrete EOF to be $\psi_k \equiv (\psi_k(\mathbf{s}_1), \dots, \psi_k(\mathbf{s}_m))'$, where ψ_k is the vector in the linear combination $a_t(k) = \psi_k' \mathbf{Z}_t$; $k = 1, \dots, m$. (Notice that for discrete EOFs, we deliberately use $\psi_k(\mathbf{s}_i)$ to refer to the i -th element of the eigenvector to distinguish it notationally from the continuous eigenfunction element $\phi_k(\mathbf{s}_i)$ evaluated at $\mathbf{s}_i$; see Section 5.3.1.) Recall we are assuming that $\text{var}(\mathbf{Z}_t)$ does not depend on time t . Then, ψ_1 is the vector that allows $\text{var}(a_t(1))$ to be maximized subject to the constraint $\psi_1' \psi_1 = 1$, and ψ_2 is the vector that maximizes $\text{var}(a_t(2))$ subject to the constraints $\psi_2' \psi_2 = 1$ and $\text{cov}(a_t(1), a_t(2)) = 0$. In general, ψ_k is the vector that maximizes $\text{var}(a_t(k))$ subject to the constraints $\psi_k' \psi_k = 1$ and $\text{cov}(a_t(k), a_t(j)) = 0$ for all $j \neq k$. This is equivalent to solving the eigen-system,

$$\mathbf{C}_Z^{(0)} \Psi = \Psi \Lambda,$$

where $\mathbf{C}_Z^{(0)} \equiv \text{var}(\mathbf{Z}_t)$ is the lag-0 spatial covariance matrix; $\Psi \equiv (\psi_1, \dots, \psi_m)$ is the $m \times m$ orthonormal matrix of eigenvectors ($\Psi' \Psi = \mathbf{I}$); $\Lambda \equiv \text{diag}(\lambda_1, \dots, \lambda_m)$ is the $m \times m$ diagonal matrix of eigenvalues decreasing

down the diagonal, $\text{var}(a_t(k)) = \lambda_k$, $k = 1, \dots, m$; and the mean of $\mathbf{Z}_t$ is assumed to be $\mathbf{0}$ without loss of generality. The solution is obtained by a symmetric decomposition of $\mathbf{C}_Z^{(0)}$, given by $\mathbf{C}_Z^{(0)} = \Psi \Lambda \Psi'$, which is a PCA formulation.

It is straightforward to show (e.g., Cohen and Jones, 1969; Buell, 1972) that if a discretization of the spatially continuous Fredholm integral equation (5.19) assumes equal areas of influence for each observation location, then such a discretization is equivalent to the spatially discrete PCA formulation of EOFs. Conversely, a PCA decomposition of irregularly spaced data (i.e., without consideration of the relative area associated with each observation location) leads to improper weighting of the significance of each element of the covariance matrix $\mathbf{C}_Z^{(0)}$ in the discretization of the spatially continuous equation (5.19). This improper weighting can give erroneous results (see Jolliffe, 2002, p. 297) and is the source of many incorrect EOF analyses in the literature. For a discussion of the effects of ignoring this distinction between the continuous and the discrete formulation, see Karl, Koscielny, and Diaz (1982).

5.3.3 Temporal Formulation

Although the most common formulation of EOFs in the literature is the spatial formulation discussed above, it is also perfectly reasonable to consider a temporal formulation of EOFs (e.g., see Preisendorfer, 1988). In this case, define the T -dimensional vectors $\mathbf{Z}(\mathbf{s}_i) \equiv (Z_1(\mathbf{s}_i), \dots, Z_T(\mathbf{s}_i))'$, for $i = 1, \dots, m$. Let the $T \times T$ covariance matrix be given by $\Gamma_Z \equiv \text{var}(\mathbf{Z}(\mathbf{s}_i))$, for all $\mathbf{s}_i \in D_s$. Then, the development of the EOF decomposition is completely analogous to the spatial formulation, and we can write the spectral decomposition as $\Gamma_Z = \Psi^{(s)} \Lambda^{(s)} \Psi^{(s)'}'$, where the superscript is used to remind us that we are decomposing a (T -dimensional) vector spatial process, $\Psi^{(s)} \equiv (\psi_1^{(s)}, \dots, \psi_T^{(s)})'$, and $\Lambda^{(s)} \equiv \text{diag}(\lambda_1^{(s)}, \dots, \lambda_T^{(s)})$. In this case, the EOFs $\{\psi_k^{(s)}: k = 1, \dots, T\}$ are T -dimensional temporal patterns. Associated with the k th EOF is a spatial field given by $a_k(\mathbf{s}_i) = (\psi_k^{(s)})' \mathbf{Z}(\mathbf{s}_i)$, for $i = 1, \dots, m$. Typically, one considers such an analysis when T , the number of time points, is much less than m , the number of spatial locations, and interest is in considering the variability associated with common temporal patterns at various spatial locations.

5.3.4 Calculation of EOFs

In this section, we focus on the calculation of the *spatial* EOFs. Analogous calculations can be performed to obtain temporal EOFs. Since EOF analysis depends on the (weighted) decomposition of a covariance matrix, $\mathbf{C}_Z^{(0)}$, in practice, we must obtain this matrix empirically. Assume that our spatio-temporal observations are given by the vectors $\mathbf{Z}_t = (Z_t(\mathbf{s}_1), \dots, Z_t(\mathbf{s}_m))'$; $t = 1, \dots, T$. Let $\hat{\mathbf{C}}_Z^{(0)}$ be the empirical temporal lag-0 covariance matrix obtained from (5.1).

Given that $\hat{\mathbf{C}}_Z^{(0)}$ is symmetric and nonnegative definite (so that all eigenvalues are greater than or equal to zero), the spectral decomposition, $\hat{\mathbf{C}}_Z^{(0)} = \hat{\Psi} \hat{\Lambda} \hat{\Psi}'$, yields the empirical eigenvectors and eigenvalues. Note that Lawley (1956) derived approximate formulas for the bias and variance of empirical eigenvalues and eigenvectors. Subsequently, von Storch and Hannoschöck (1985) extended Lawley's results by allowing for possible temporal dependence in $\{\mathbf{Z}_t : t = 1, \dots, T\}$. Based on theory and Monte Carlo simulation, they found that the empirical eigenvalue $\hat{\lambda}_k$ is biased relative to λ_k . This bias is positive for the larger λ_k 's and negative for the smaller λ_k 's. Furthermore, it has been shown that the sampling error associated with the covariance matrix $\hat{\mathbf{C}}_Z^{(0)}$ leads to numerical instability in the EOFs, $\hat{\Psi}$ (e.g., Gray, 1981; North et al., 1982). This has led to sampling-based selection strategies for the level of truncation, p_α . Many of these strategies are described in Preisendorfer (1988).

We note that in the presence of white-noise measurement error, the EOFs corresponding to the true, non-noisy process (say, Y) are the same as for the data (say, $Z = Y + \text{noise}$), but the eigenvalues are different. In addition, the "amplitude" time series for both the data (Z) and true process (Y) have the same mean (zero), but the variance of the true process' "amplitude" time series is less. Thus, if white-noise measurement error is present, and EOFs are calculated based on the data, then the EOF spatial patterns are not affected; however, the variance accounted for by each pattern must be adjusted accordingly, by subtracting the estimated white-noise variance from each estimated eigenvalue.

Calculation When $m > T$

Consider the situation in which the number of spatial locations m is larger than the number of time periods T , and we would like to calculate the EOFs associated with the spatial covariance matrix $\hat{\mathbf{C}}_Z^{(0)}$ given by (5.1). In this case, the $m \times m$ covariance matrix is not positive-definite, since it only has $T - 1$ nonzero eigenvalues. Thus, there are a maximum of $T - 1$ unique EOFs available for consideration. In many cases, $m \gg T$, and it is computationally inefficient to perform the spectral decomposition on $\hat{\mathbf{C}}_Z^{(0)}$. However, we can make use of a convenient matrix-algebra result (e.g., von Storch and Hannoschöck, 1984). Consider an $m \times T$ matrix $\mathbf{W}$. It can be shown that the nonzero eigenvalues of the $m \times m$ matrix $\mathbf{W}\mathbf{W}'$ are the same as the nonzero eigenvalues of the $T \times T$ matrix $\mathbf{W}'\mathbf{W}$. Furthermore, it can be shown that if ξ_i is an eigenvector of $\mathbf{W}'\mathbf{W}$ corresponding to nonzero eigenvalue λ_i , then $\psi_i \equiv \mathbf{W}\xi_i / \sqrt{\xi_i'\mathbf{W}'\mathbf{W}\xi_i}$ is the eigenvector of $\mathbf{W}\mathbf{W}'$ associated with λ_i .

For a spatio-temporal dataset that does not have missing values, we can define the $m \times T$ matrix of data, $\tilde{\mathbf{Z}} \equiv (\tilde{\mathbf{Z}}_1, \dots, \tilde{\mathbf{Z}}_T)$, where $\tilde{\mathbf{Z}}_t \equiv \mathbf{Z}_t - \bar{\mu}_Z$ and $\bar{\mu}_Z$ is given by (5.2). Now, define the $T \times T$ matrix, $\mathbf{A} \equiv \tilde{\mathbf{Z}}'\tilde{\mathbf{Z}}$; then there are a maximum of $T - 1$ nonzero eigenvalues associated with $\mathbf{A}$. Next, calculate the spectral decomposition, $\mathbf{A} = \hat{\Xi} \hat{\Lambda} \hat{\Xi}'$, where $\hat{\Xi} \equiv (\hat{\xi}_1, \dots, \hat{\xi}_T)$. Then, for each

nonzero $\hat{\lambda}_i$, one obtains the associated EOF of $\hat{\mathbf{C}}_Z^{(0)}$ by using the matrix-algebra result above: $\hat{\psi}_i \equiv \tilde{\mathbf{Z}}\hat{\xi}_i / \sqrt{\hat{\xi}_i' \tilde{\mathbf{Z}}' \tilde{\mathbf{Z}} \hat{\xi}_i}$; $i = 1, \dots, T - 1$.

Example: Consider the monthly SST anomaly dataset $\{\mathbf{Z}_t : t = 1, \dots, 396\}$ for the 33-year period from January 1970 to December 2002, discussed in Section 5.1. Figure 5.17 shows the first and second EOFs and the associated “amplitude” time series for the 33-year period. Now, EOF 1 and EOF 2 account for 38.8% and 9.3% of the variation in the dataset, respectively. Note that the spatial pattern for the first EOF looks very much like the El Niño pattern shown in Figure 5.4. Indeed, the associated “amplitude” time series shows peaks corresponding to the peaks in the two El Niño time series shown in Figure 5.2. The first EOF accounts for a remarkably large percentage of the variation over the 30-plus years of data, illustrating just how important the El Niño phenomenon is in the tropical Pacific. The second EOF also shows an El Niño-like “signature,” but it is much more constrained to the eastern portion of the tropical Pacific region. In contrast, Figure 5.18 shows the third and fourth EOFs and associated “amplitude” time series; these account for 8.5% and 4.2% of the variation, respectively.

Although La Niña signatures appear in the central portion of the equatorial Pacific, there are other spatial regions showing large (in magnitude) EOF coefficients (sometimes called loadings) as well. This illustrates an important point about EOF analysis, namely the constraint of orthogonality limits the physical interpretability of all but the first EOF. However, this example also illustrates the incredible dimension-reduction capability of EOF analysis, something we use when fitting a BHM to the SST anomaly dataset in Section 9.1. In this case, the first four EOFs account for over 60% of the variation of a spatial vector that has $m = 2,261$ elements (i.e., the number of ocean pixels in the domain of interest). Figure 5.19 shows the cumulative percentage of variation accounted for by the first 100 EOFs; the first 100 EOFs account for over 97% of the SST-anomaly variation.

5.4 EXTENSIONS OF EOF ANALYSIS

The eigenvalue/eigenvector decomposition underlying EOF analysis can be applied to a variety of other situations involving spatio-temporal data. In most cases, these can be thought of as summarizing in some manner an augmented data matrix. We briefly discuss three of these cases below. For the interested reader, the Bibliographic Notes in Section 5.8 contain references that give considerably more detail.

5.4.1 Complex (Hilbert) EOFs

One advantage of the EOF approach described in Section 5.3 is that the complicated variability of spatio-temporal data can be compressed into a relatively

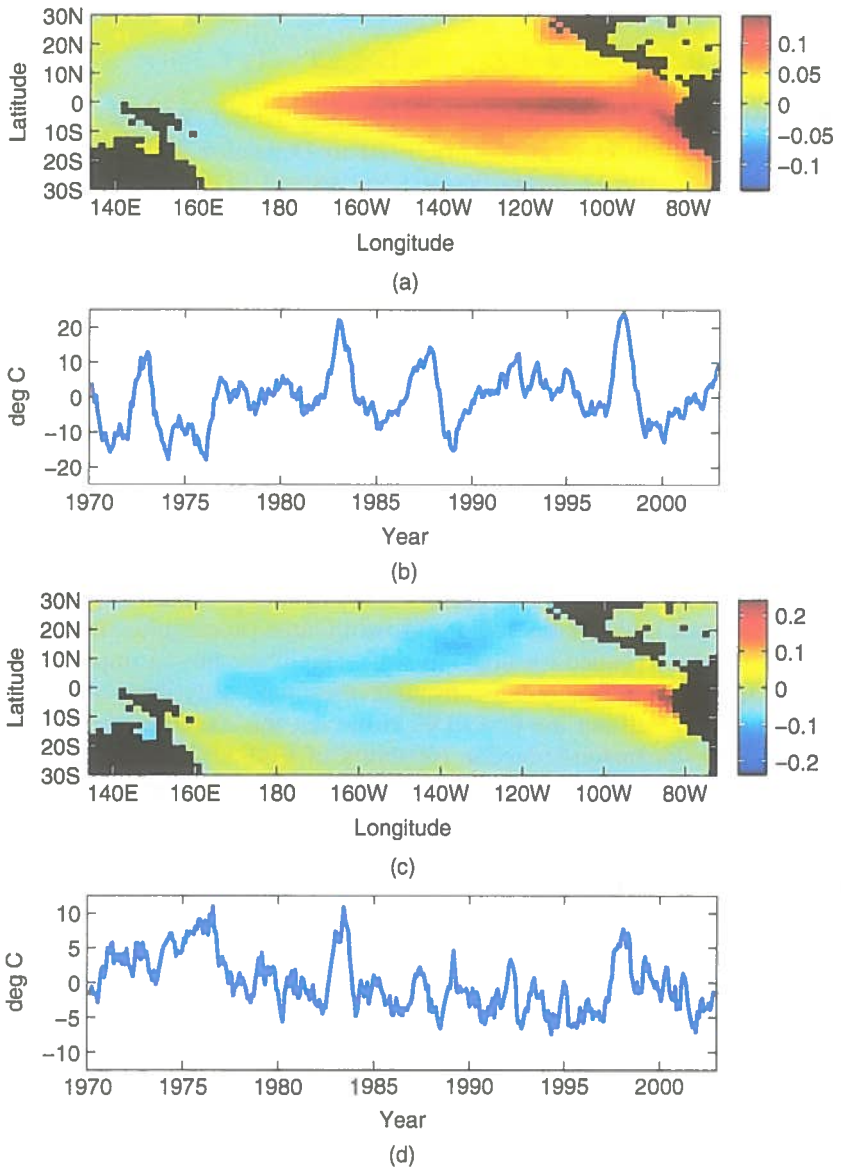


Figure 5.17 (a) First EOF for SST anomaly data. (b) Associated amplitude time series for the first EOF, which accounts for 38.8% of the variation in the monthly anomalies from January 1970 through December 2002. (c) Second EOF for SST anomaly data. (d) Associated amplitude time series for the second EOF, which accounts for 9.3% of the variation in these data. Note that the EOFs are unitless in these plots, and the amplitude time series are in the same units as the data, °C.

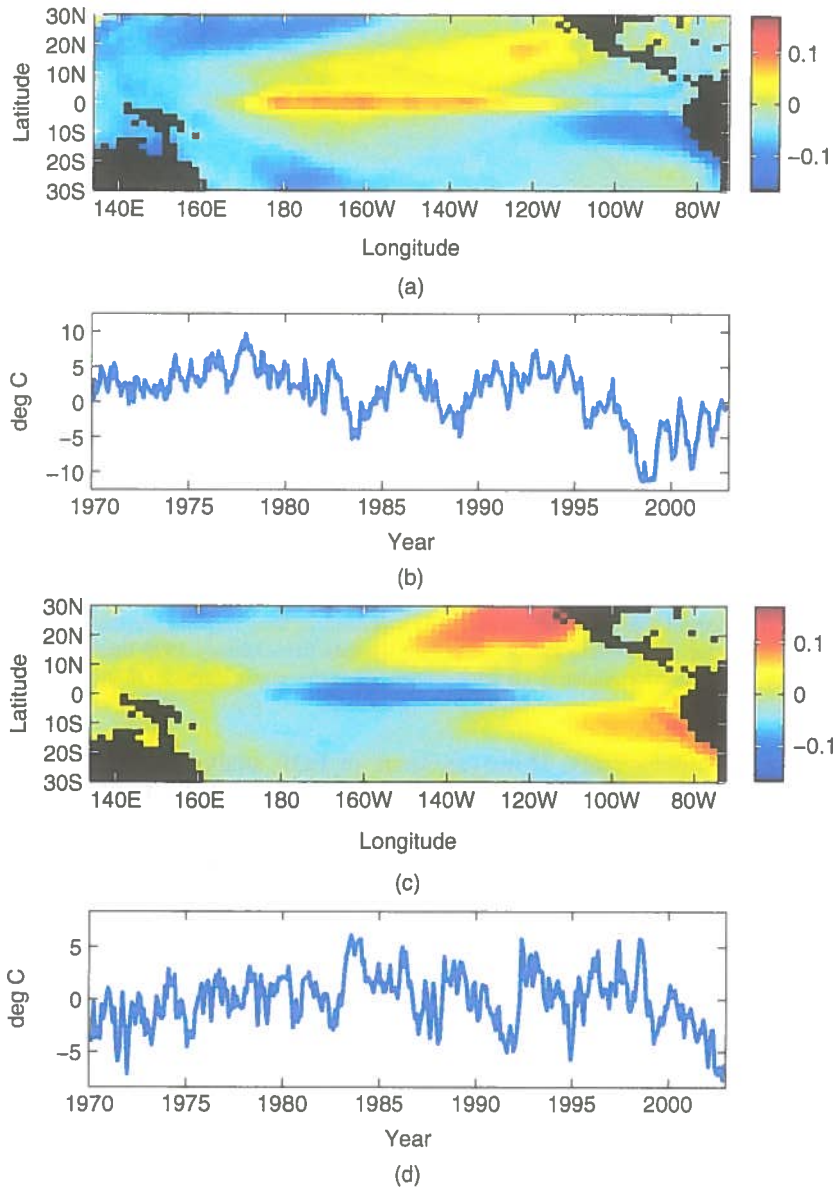


Figure 5.18 (a) Third EOF for SST anomaly data. (b) Associated amplitude time series for the third EOF, which accounts for 8.5% of the variation in the monthly anomalies from January 1970 through December 2002. (c) Fourth EOF for SST anomaly data. (d) Associated amplitude time series for the fourth EOF, which accounts for 4.2% of the variation in these data. Units are the same as in Figure 5.17.

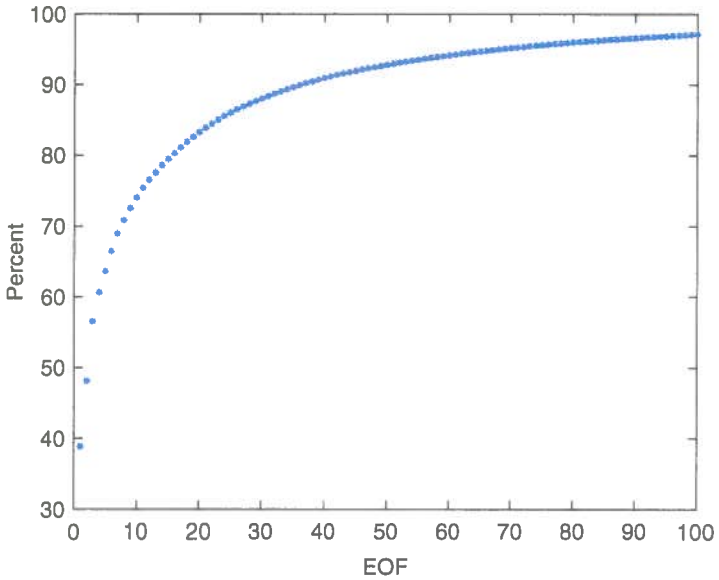


Figure 5.19 Cumulative variance, expressed as a percent of total variance, accounted for by the first 100 EOFs for the monthly Pacific SST anomaly data from January 1970 through December 2002. The y axis shows cumulative variance (percent) and the x axis shows EOF index.

small set of eigenvectors. However, such an EOF analysis only detects spatial structures that do not propagate in time. To extend the EOF analysis to the study of spatial structures that can propagate in time, Wallace and Dickinson (1972) developed *complex principal component* analysis in the frequency domain. The technique involves the computation of complex eigenvectors from cross-spectral matrices. The limitation of this technique is that it only gives the decomposition for individual (i.e., very narrow) frequency bands. Consequently, if the power of a phenomenon is spread over a wide frequency band (as is generally the case with environmental phenomena), then several EOF spatial maps (one for each spectral estimate) are needed to evaluate the phenomenon, and this can complicate the interpretation.

Complex empirical orthogonal function (CEOF) analysis, or *Hilbert EOF (HEOF) analysis*, in the time domain, was developed as an alternative to the frequency-domain approach described above (Rasmusson, Arkin, and Chen, 1981). This method differs from the frequency-domain approach in that Hilbert transforms are used to shift the time series of the data at each location by a quarter of a cycle. That is, for a data vector $\mathbf{Z}_t = (Z_t(s_1), \dots, Z_t(s_m))'$, let

$$\tilde{\mathbf{Z}}_t \equiv \mathbf{Z}_t + i\mathbf{Z}_t^H, \quad (5.24)$$

where $i = \sqrt{-1}$ and $\mathbf{Z}_t^H$ is the *Hilbert transformed* vector defined by (e.g., von Storch and Zwiers, 1999, Section 16.2.2):

$$\mathbf{Z}_t^H \equiv \sum_{j=0}^{\infty} \frac{2}{(2j+1)\pi} (\mathbf{Z}_{t+2j+1} - \mathbf{Z}_{t-2j+1}), \quad (5.25)$$

for $t = \dots, -1, 0, 1, \dots$.

The original data and its Hilbert transform allow the examination of propagating disturbances. Specifically, one simply performs the EOF analysis (described in Section 5.3) on the empirical covariance matrix of the complex augmented data, $\{\tilde{\mathbf{Z}}_t : t = 1, \dots, T\}$. Since this covariance matrix is Hermitian, it has complex eigenvectors and real, nonnegative eigenvalues (e.g., von Storch and Zwiers, 1999, p. 359). Barnett (1983) and Horel (1984) give excellent discussions of some of the theoretical and practical issues related to such analyses.

If one can compute the Hilbert transform (5.25), then the EOF analysis of $\{\tilde{\mathbf{Z}}_t\}$ proceeds directly after calculating its empirical covariance matrix and performing the eigenvector/eigenvalue decomposition, exactly as in the usual EOF analysis. One should be sure to take the complex conjugate of $(\mathbf{Z}_{t-\tau} - \bar{\mu}_Z)$ in (5.1) when calculating the (lagged) empirical covariance matrix. In practice, finite-sample Hilbert transforms can be computed fairly easily, as discussed in Barnett (1983) and von Storch and Zwiers (1999, Section 16.2.4). Of course, the eigenvectors (EOFs) and the principal component (“amplitude”) time series are both complex-valued. Thus, one typically has to develop some intuition regarding the interpretation of results. The following example is provided to illustrate a basic HEOF analysis.

Example: Consider the monthly SST anomaly dataset $\{\mathbf{Z}_t : t = 1, \dots, 396\}$ for the 33-year period from January 1970 to December 2002, discussed in Section 5.1. Then, $\{\tilde{\mathbf{Z}}_t\}$ is constructed as in (5.24), where $\{\mathbf{Z}_t^H\}$ defined by (5.25) is calculated in the manner suggested by von Storch and Zwiers (1999, Section 16.2.4). The first complex EOF accounts for 42.4% of the variation in this dataset. (Since the augmented covariance matrix is Hermitian, recall that the eigenvalues are real and thus this percentage is calculated by simply dividing the largest eigenvalue by the sum of all eigenvalues.) Image plots of the real and imaginary part of the first complex EOF are shown in the top and bottom panels of Figure 5.20, respectively. Figure 5.21 shows the corresponding real and imaginary “amplitude” time series for the first complex EOF in the top panel. The bottom panel shows the associated *phase* through time, given by $\arctan\{Im[a_t(1)]/Re[a_t(1)]\}$, where $a_t(1)$ is the complex-valued “amplitude” time series associated with the first complex EOF, and $Im[\cdot]$ and $Re[\cdot]$ correspond to the imaginary and real components, respectively. Note that the real part of the first complex EOF resembles the spatial pattern of the El

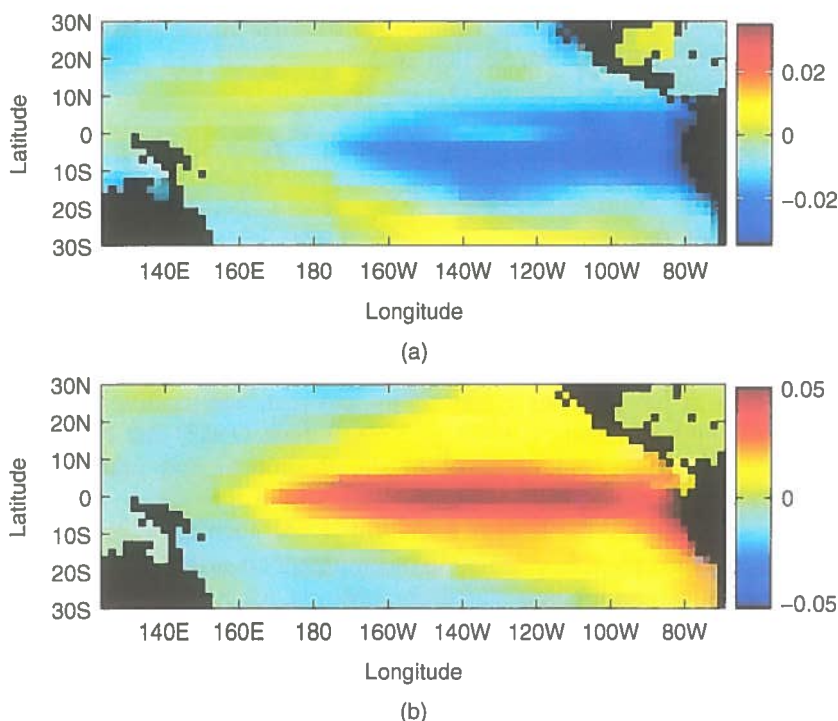


Figure 5.20 The first complex (Hilbert) EOF from the monthly Pacific SST anomaly data from January 1970 to December 2002. (a) The real part of the first complex EOF. (b) The imaginary part of the first complex EOF. This complex EOF accounts for approximately 42.4% of the total variability of the Hilbert transform-augmented Pacific SST anomalies [see (5.24)] over this time period, as derived from the largest eigenvalue of the associated covariance matrix. The EOFs are unitless.

Niño signal when the real part of $a_t(1)$ is negative, and somewhat resembles spatial pattern of the La Niña signal when the real part of $a_t(1)$ is positive. Similarly, the pattern of the corresponding imaginary part of the first complex EOF resembles a narrower (in a latitudinal sense) El Niño pattern when the imaginary part of $a_t(1)$ is positive, and it resembles a La Niña pattern when the imaginary part of $a_t(1)$ is negative. In essence, these patterns, and the associated time series, show an idealized and filtered view of the ENSO phenomena.

Let ψ_1 be the first complex EOF; then our CEOF analysis suggests that one cycle of ENSO follows roughly the sequence: $Re\{\psi_1\}$, $-Im\{\psi_1\}$, $-Re\{\psi_1\}$, $Im\{\psi_1\}$. Graphically, this corresponds to the top panel in Figure 5.20, followed 1/4 cycle later by the negative of the bottom panel of Figure 5.20, followed by the negative of the top panel, followed by the bottom panel.

The real and imaginary parts of the time series shown in Figure 5.21 are roughly a quarter cycle out of phase, as expected, since the imaginary part is the Hilbert transform of the real part (von Storch and Zwiers, 1999, Sec. 16.3.14).

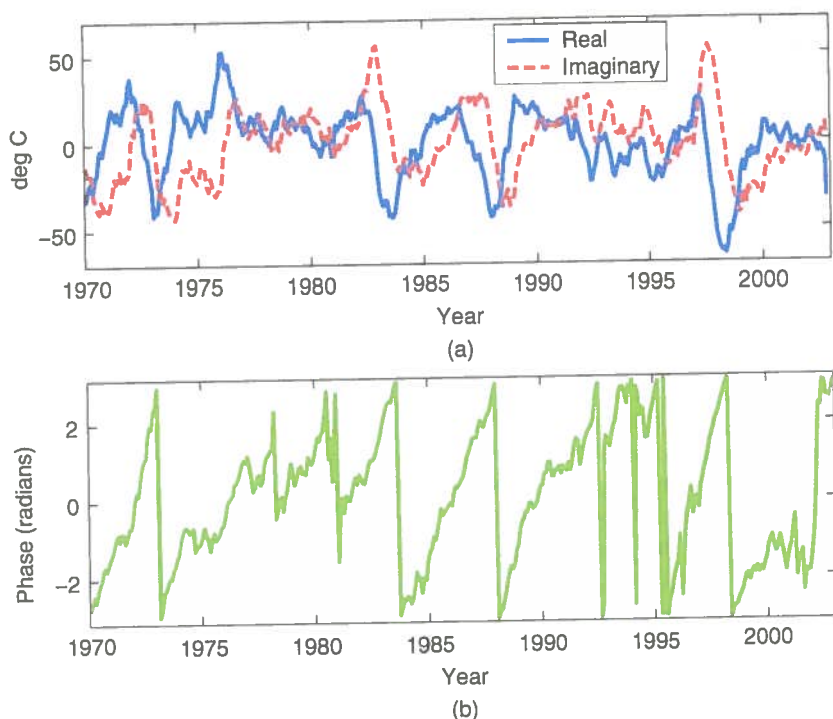


Figure 5.21 The first complex (Hilbert) EOF amplitude time series corresponding to the CEOFs shown in Figure 5.20. (a) The real (blue) and imaginary (red) portions of the amplitude time series. (b) The phase (green) associated with the real and imaginary portions (i.e., the arctan of the imaginary part divided by the real part of the complex-valued time series).

More importantly, the bottom panel of Figure 5.21 shows the associated phase in radians of the ENSO cycle discussed above. Each “sawtooth” in the phase plot represents a filtered ENSO oscillation. Clearly, the period of these cycles varies considerably throughout the data record.

As discussed in von Storch and Zwiers (1999, Sec. 16.3.12), the interpretation of complex EOF analysis can be difficult and is based on some tenuous assumptions. They wisely state: “Prudence is clearly advisable” (von Storch and Zwiers, 1999, p. 363).

5.4.2 Multivariate EOF Analysis

There is often interest in the simultaneous analysis of two or more spatio-temporal datasets. Kutzbach (1967) used a form of EOF analysis that simultaneously considered several meteorological variables at many spatial locations and times, and Preisendorfer (1988) discussed this approach at length. A brief description of the basic idea behind the multivariate EOF analysis follows.

Consider two datasets observed over time at the same spatial locations; that is, consider $Z_t(s_i)$ and $X_t(s_i)$, where $i = 1, \dots, m$; $t = 1, \dots, T$. We write the augmented vector as $\mathbf{W}_t \equiv (\mathbf{Z}'_t, \mathbf{X}'_t)'$, where $\mathbf{Z}_t \equiv (Z_t(s_1), \dots, Z_t(s_m))'$ and $\mathbf{X}_t \equiv (X_t(s_1), \dots, X_t(s_m))'$. Denote the covariance matrix of $\mathbf{W}_t$ by $\mathbf{C}_W^{(0)}$. This matrix includes off-diagonal submatrices that represent the covariance between $\mathbf{Z}_t$ and $\mathbf{X}_t$. As shown in Preisendorfer (1988, pp. 161–162), the EOFs can then be obtained in the conventional manner from the eigen-decomposition of $\mathbf{C}_W^{(0)}$; that is, $\mathbf{C}_W^{(0)} = \Psi_W \Lambda_W \Psi_W'$, where the columns of Ψ_W are the eigenvectors (i.e., EOFs) and Λ_W is a diagonal matrix containing the eigenvalues of $\mathbf{C}_W^{(0)}$. Then, the first m elements of the k th eigenvector correspond to the portion of the k th EOF for the process $\{\mathbf{Z}_t : t = 1, \dots, T\}$, and the last m elements correspond to the portion of the k th EOF for the process $\{\mathbf{X}_t : t = 1, \dots, T\}$.

Theoretically, there is no limit to the number of processes that could be considered simultaneously. However, there is a practical limitation to this procedure, since the covariance matrix can easily become very large if the number of observation locations or variables increases. Bretherton, Smith, and Wallace (1992) compare this approach to other multivariate methods such as Canonical Correlation Analysis (see Section 5.6) and find that, in some cases, the multivariate EOF analysis has large biases and does not perform well when the signal-to-noise ratio is small. Calculation of multivariate EOFs proceeds as with single-process EOFs, by starting with the empirical covariance matrix $\hat{\mathbf{C}}_W^{(0)}$.

5.4.3 Extended EOF Analysis

Extended EOFs (e.g., Weare and Nasstrom, 1982) are simply multivariate EOFs in which the additional variables are lagged versions of the same process. For example, we could let the augmented data vector be $\mathbf{W}_t = (\mathbf{Z}'_t, \mathbf{Z}'_{t-\tau})'$, for $\tau = 1, 2, \dots$. In this case, if temporal invariance is assumed, then the diagonal submatrices of $\mathbf{C}_W^{(0)}$ are equivalent, and the off-diagonal submatrices are just the lag- τ covariance matrices $\mathbf{C}_Z^{(\tau)}$. In this way, we can examine the propagation of EOF spatial patterns in time by noting that the first m coefficients of a particular eigenvector correspond to the lag-zero representation of that EOF, and the next m coefficients correspond to the lag- τ representation of the same EOF. This approach is closely linked with time-lagged Canonical Correlation Analysis discussed below in Section 5.6.

Example: Consider the monthly SST anomaly dataset $\{\mathbf{Z}_t : t = 1, \dots, 396\}$ for the 33-year period from January 1970 through December 2002, discussed in Section 5.1. In this case, we form the augmented data vector with the SST anomaly data at time t and at time $t - 6$ (i.e., $\tau = 6$ months) throughout the data record. The extended EOF patterns for the first eigenvector are shown in Figure 5.22. The top panel shows the portion of the first eigenvector corresponding to the SST anomalies at time t , while the lower panel shows the portion of the same eigenvector corresponding to the anomalies at time $t - 6$.

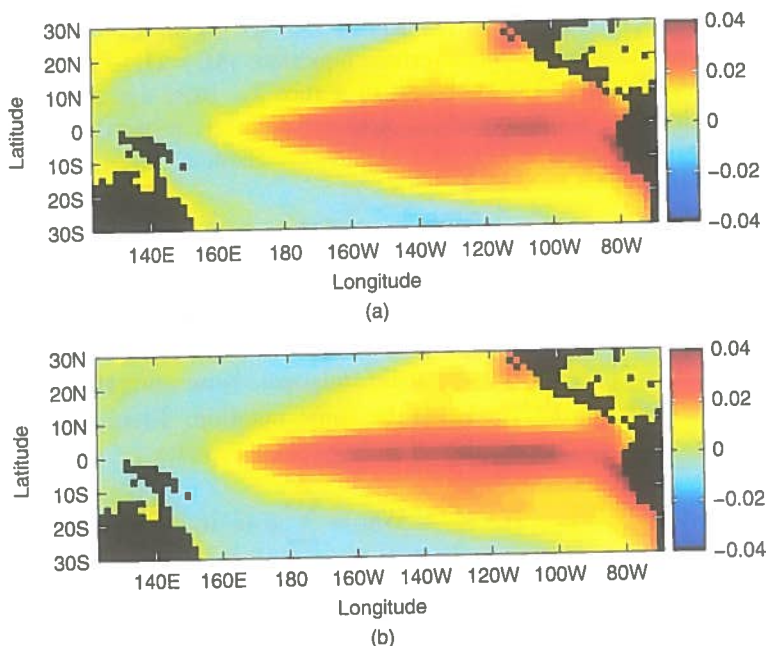


Figure 5.22 The first extended EOF of the monthly Pacific SST anomaly data from January 1970 to December 2002. (a) The portion of the eigenvector associated with the current time t and (b) the portion of the eigenvector associated with time-lagged $t - 6$. This extended EOF accounts for 31% of the variation in the augmented data vector. Note that these eigenvectors are unitless.

This first extended EOF accounts for 31% of the variation in the augmented data vector. The ENSO pattern is evident in both plots in Figure 5.22, suggesting that there is substantial persistence across 6 months. Nonetheless, there are differences in the spatial pattern of these EOFs, indicating that there is variability in the spatial pattern throughout the ENSO cycle.

5.5 PRINCIPAL OSCILLATION PATTERNS (POPs)

Recall from Section 3.2.1 that a spectral decomposition of the *propagator matrix* for a first-order dynamical system elucidates many features of its underlying dynamics. In essence, Principal Oscillation Pattern (POP) analysis is just the spectral decomposition of the operator for a spatio-temporal process expressed as a first-order dynamical system.

To gain insight into the interpretation of the POPs in space and time, we consider an idealized first-order discrete linear system, as discussed in Section 3.2.1:

$$\mathbf{Z}_t = \mathbf{M}\mathbf{Z}_{t-1}, \quad (5.26)$$

where $\mathbf{Z}_t \equiv (Z_t(s_1), \dots, Z_t(s_m))'$. Recall that for an $m \times m$ full-rank non symmetric matrix $\mathbf{M}$, the characteristic equation, $|\mathbf{M} - \lambda \mathbf{I}| = 0$, has m (possibly complex) roots. Corresponding to a root λ_k , there are two vectors $\mathbf{w}_k$ and $\mathbf{v}_k$, called the “right” and “left” *singular vectors*, respectively, such that $\mathbf{M}\mathbf{w}_k = \lambda_k \mathbf{w}_k$ and $\mathbf{M}'\mathbf{v}_k = \lambda_k \mathbf{v}_k$; $k = 1, \dots, m$. Then,

$$\mathbf{M} = \sum_{k=1}^m \frac{\lambda_k}{d_k} \mathbf{w}_k \mathbf{v}_k', \quad (5.27)$$

where $d_k \equiv \mathbf{v}_k' \mathbf{w}_k > 0$. Recall from Section 3.2.1 that $\mathbf{W}\mathbf{V}' = \mathbf{D}$, where $\mathbf{W} \equiv (\mathbf{w}_1, \dots, \mathbf{w}_m)$, $\mathbf{V} \equiv (\mathbf{v}_1, \dots, \mathbf{v}_m)$, and $\mathbf{D}$ is a diagonal matrix with the elements $\{d_k : k = 1, \dots, m\}$ on the main diagonal. Now normalize $\mathbf{w}_k$ and $\mathbf{v}_k$ by dividing each by $d_k^{1/2}$ and use the same notation. Then, $\mathbf{W}\mathbf{V}' = \mathbf{I}$, and $\mathbf{M} = \mathbf{W}\mathbf{\Lambda}\mathbf{V}' = \sum_{k=1}^m \lambda_k \mathbf{w}_k \mathbf{v}_k'$. Consequently, we can write

$$\mathbf{Z}_t = \mathbf{W}\mathbf{V}'\mathbf{Z}_t = \mathbf{W}\mathbf{a}_t = \sum_{k=1}^m a_t(k) \mathbf{w}_k, \quad (5.28)$$

where

$$\mathbf{a}_t \equiv \mathbf{V}'\mathbf{Z}_t \quad (5.29)$$

and $\mathbf{a}_t \equiv (a_t(1), \dots, a_t(m))'$. The essence of POP analysis is the decomposition represented by (5.27) and (5.29). The right singular vectors $\{\mathbf{w}_k : k = 1, \dots, m\}$ are called *principal oscillation patterns* and, although they constitute a linear basis, they are not orthonormal. Each of these vectors does represent a spatial field, as each component of the vector is spatially referenced. The elements of the time series $\{a_t(k) : t = 1, 2, \dots\}$ are known as POP coefficients. The left singular vectors $\{\mathbf{v}_k : k = 1, \dots, m\}$ are called *adjoint bases*. We note that, although the basis vectors $\{\mathbf{w}_k\}$ are not orthogonal with themselves, any $\mathbf{w}_k$ is orthogonal to $\mathbf{v}_j$; $j \neq k$.

The right singular vectors $\{\mathbf{w}_k : k = 1, \dots, m\}$ of the matrix $\mathbf{M}$ are sometimes referred to as the *normal modes* of the first-order dynamical system. Because $\mathbf{M}$ is not in general symmetric, some or all of its eigenvalues and eigenvectors are complex, as discussed in Section 3.2.1. Furthermore, because $\mathbf{M}$ is real, the complex conjugates, $\bar{\lambda}_k$ and $\bar{\mathbf{w}}_k$, also satisfy the singular-vector equation, $\mathbf{M}\bar{\mathbf{w}}_k = \bar{\lambda}_k \bar{\mathbf{w}}_k$.

Now, writing (5.26) in terms of its decompositions (5.27) and (5.28) and pre-multiplying by $\mathbf{V}'$, we obtain from (5.29)

$$\mathbf{a}_t = \mathbf{V}'\mathbf{W}\mathbf{\Lambda}\mathbf{V}'\mathbf{Z}_{t-1} = \mathbf{\Lambda}\mathbf{a}_{t-1}, \quad (5.30)$$

or

$$a_t(k) = \lambda_k a_{t-1}(k), \quad k = 1, \dots, m, \quad (5.31)$$

which is a collection of m first-order homogeneous difference equations, each with solution $a_t(k) = (\lambda_k)^t$ (assuming $a_0(k) = 1$), as described in Section 3.2.

If λ_k is complex, then $\lambda_k \equiv \lambda_k^R + i\lambda_k^I$ (recall $i \equiv \sqrt{-1}$), which can be written in polar form as $\lambda_k^R = \gamma_k \cos(\phi_k)$ and $\lambda_k^I = \gamma_k \sin(\phi_k)$. Then, $\lambda_k = \gamma_k e^{i\phi_k}$, where $\gamma_k = |\lambda_k| = \{(\lambda_k^R)^2 + (\lambda_k^I)^2\}^{1/2}$ and $\phi_k = \arctan(\lambda_k^I/\lambda_k^R)$. Thus, the coefficients evolve according to, $a_i(k) = \gamma_k^i e^{i\phi_k i}$. Under the conditions, $|\lambda_k| \leq 1$; $k = 1, \dots, m$, $a_i(k)$ evolves as a damped spiral in the complex plane with a characteristic damping rate γ_k and frequency ϕ_k , for $k = 1, \dots, m$. Notice that here ϕ_k is an *angle* in the complex plane and should not be confused with the basis functions and their associated vectors that we use in EOF representations of spatio-temporal processes (e.g., Section 5.3).

We now decompose the right singular vector (i.e., normal mode) $\mathbf{w}_k$ as the sum of a real and an imaginary term: $\mathbf{w}_k = \mathbf{w}_k^R + i\mathbf{w}_k^I$. Note that for $\mathbf{M}$ real, the normal modes occur in complex conjugate pairs if they are complex. Hence, the general evolution of a damped normal mode (i.e., $\gamma_k \leq 1$) can be described in a two-dimensional subspace spanned by $\mathbf{w}_k^R$ and $\mathbf{w}_k^I$ [see, e.g., von Storch et al. (1995)]; it occurs in the succession,

$$\dots \rightarrow \mathbf{w}_k^R \rightarrow -\mathbf{w}_k^I \rightarrow -\mathbf{w}_k^R \rightarrow \mathbf{w}_k^I \rightarrow \mathbf{w}_k^R \rightarrow \dots, \quad (5.32)$$

with a period of $2\pi/\phi_k$; and each stage in (5.32) occurs a quarter of a cycle apart. Note the similarity of this to the complex (Hilbert) EOF example shown in Section 5.4.1. In the case of POPs, the time τ_k needed to reduce an initial amplitude $a_0(k)$ to $a_0(k)/e$ is referred to as the *e-folding time* and is given by $\tau_k \equiv -1/\ln(\gamma_k)$.

5.5.1 Calculation of POPs

The previous section emphasized the physical motivation and interpretation behind a POP analysis. If the system were *deterministic*, then the normal-mode approach leading to (5.32) would be directly interpretable. However, data come with uncertainty, and we need to account for it. Thus, we assume that the data follow a first-order vector autoregressive process (i.e., VAR(1), as discussed in Section 3.4.7):

$$\mathbf{Z}_t = \mathbf{M}\mathbf{Z}_{t-1} + \boldsymbol{\eta}_t, \quad (5.33)$$

where $\boldsymbol{\eta}_t$ is an m -dimensional additive error vector such that $E(\boldsymbol{\eta}_t) = \mathbf{0}$, $E(\boldsymbol{\eta}_t \mathbf{Z}'_{t-1}) = \mathbf{0}$, and $E(\boldsymbol{\eta}_t \boldsymbol{\eta}'_r) = \mathbf{C}_\eta^{(0)}$ when $t = r$, and $\mathbf{0}$ otherwise. As discussed in Section 3.4.7, under second-order stationarity, $\mathbf{M} = \mathbf{C}_Z^{(1)} (\mathbf{C}_Z^{(0)})^{-1}$, where $\mathbf{C}_Z^{(\tau)}$ is the lag- τ cross-covariance matrix for $\{\mathbf{Z}_t\}$.

In order to perform a POP analysis in practice, the propagator matrix $\mathbf{M}$ in (5.33) must be estimated. As discussed in Section 3.4.7, a method-of-moments estimator for $\mathbf{M}$ is $\hat{\mathbf{M}} = \hat{\mathbf{C}}_Z^{(1)} (\hat{\mathbf{C}}_Z^{(0)})^{-1}$, where $\hat{\mathbf{C}}_Z^{(\tau)}$; $\tau = 0, 1$, are given by (5.1). An eigenvector/eigenvalue decomposition of $\hat{\mathbf{M}}$ then gives estimated right singular vectors $\{\hat{\mathbf{w}}_k: k = 1, \dots, m\}$ and estimated eigenvalues $\{\hat{\lambda}_k: k = 1, \dots, m\}$. The adjoint-vector (left-singular-vector) estimates can

be obtained from $\hat{\mathbf{V}}' = \hat{\mathbf{W}}^{-1}$, and the POP-coefficient estimates can then be obtained from $\hat{\mathbf{a}}_t = \hat{\mathbf{V}}' \mathbf{Z}_t$; see (5.29). Similarly, estimates of $\{\gamma_k\}$ and $\{\phi_k\}$ can be obtained from the estimates $\{\hat{\lambda}_k\}$.

The estimated eigenvectors $\hat{\mathbf{W}}$ are sometimes referred to as *empirical* normal modes, analogous to the deterministic decomposition. It is then *assumed* that these empirical normal modes behave as one would expect the deterministic normal modes to behave. For example, for damped empirical modes (i.e., $\hat{\gamma}_k \leq 1$, where $\hat{\gamma}_k \equiv |\hat{\lambda}_k|$) one might expect the succession (5.32). However, in the presence of noise, this relationship may not hold. To check if this relationship is valid in practice, a cross-spectral analysis is often performed between the estimates of the real component $\hat{a}_t^R(k)$ and the imaginary component $\hat{a}_t^I(k)$ of $\hat{a}_t(k)$ which, according to the deterministic analysis, should vary coherently with a frequency ϕ_j and phase lag $\pi/2$, with $\hat{a}_t^R(k)$ lagging $\hat{a}_t^I(k)$. If this is not observed, then the interpretation based on (5.32) is not valid.

Although POPs can be used for both prognostic and diagnostic purposes, from a statistical perspective they are probably best considered as diagnostic, exploratory tools. As diagnostics, POPs are used to examine the oscillation properties and spatial structure of spatio-temporal dynamical processes. In this case, one looks at the estimated frequencies $\{\hat{\phi}_k\}$, amplitudes $\{\hat{\gamma}_k\}$, and e-folding times $\{\hat{\tau}_k\}$, as well as the right singular vectors (POPs) $\{\hat{\mathbf{w}}_k\}$ and amplitude time series $\{\hat{a}_t(k)\}$. These quantities give insight into the interpretation of the empirical normal modes. Often, the data are initially filtered (usually in time) to focus on a particular phenomenon of interest, assuming the appropriate frequency band in which to filter is known *a priori*. In some cases, the data are projected onto a reduced set of basis functions (e.g., EOFs) and the POP analysis is performed in the reduced-dimensional space. This is in the spirit of dimension reduction presented in Section 7.2.6, and we demonstrate the idea in the following example.

Example: Consider the monthly SST anomaly dataset $\{\mathbf{Z}_t\}$ from January 1970 to December 2002, discussed in Section 5.1. For diagnostic purposes, assume a VAR(1) structure,

$$\mathbf{Z}_t = \mathbf{M}\mathbf{Z}_{t-1} + \boldsymbol{\eta}_t, \quad (5.34)$$

with the same assumptions as given below (5.33). A POP analysis considers the decomposition of the estimated propagator matrix $\hat{\mathbf{M}}$, however there are $T = 33 \times 12 = 396$ time points (months) and $m = 2,261$ ocean pixels (spatial locations) in the dataset. Thus, the method-of-moments estimate, $\hat{\mathbf{M}} \equiv \hat{\mathbf{C}}_Z^{(1)}(\hat{\mathbf{C}}_Z^{(0)})^{-1}$, whose components are given by (5.1), is not well defined (the $2,261 \times 2,261$ matrix, $\hat{\mathbf{C}}_Z^{(0)}$, has rank $T - 1 = 395$). Yet, as we have seen, most of the variability in the monthly SST anomalies can be accounted for by a relatively small number of EOFs. Thus, let us assume we can approximate the SST anomaly data by projecting onto the first p_α EOFs, given by the $m \times p_\alpha$ matrix $\boldsymbol{\Psi}$: From Section 5.3.2, $\mathbf{Z}_t \simeq \boldsymbol{\Psi} \cdot (a_t(1), \dots, a_t(p_\alpha))'$, where

‘.’ is used here to emphasize matrix–vector multiplication. Hence, *define* the (dimension-reduced) p_α -variate vector,

$$\alpha_t \equiv \Psi'Z_t.$$

Then, ignoring the truncation error and using orthogonality of the EOFs, we can rewrite (5.34) as

$$\alpha_t = \Psi'M\Psi\alpha_{t-1} + \Psi'\eta_t \equiv M_\alpha\alpha_{t-1} + \eta_{\alpha,t}, \tag{5.35}$$

where $M_\alpha \equiv \Psi'M\Psi$ and $\eta_{\alpha,t} \equiv \Psi'\eta_t$. Now, the POP analysis can be performed on the p_α -dimensional vectors, $\{\alpha_t : t = 1, \dots, T\}$, leading to the estimates $\hat{C}_\alpha^{(r)}$ and hence to $\hat{M}_\alpha \equiv \hat{C}_\alpha^{(1)}(\hat{C}_\alpha^{(0)})^{-1}$. Notice that there is no concern about the singularity of the lag-0 covariance matrix when $p_\alpha < T$. Furthermore, interpretations of the POPs can still be considered in physical space. To see this, note that the POP analysis yields the singular value decomposition, $\hat{M}_\alpha = \hat{W}_\alpha \hat{\Lambda}_\alpha \hat{V}_\alpha'$, and thus we can define $\hat{M} \equiv \Psi \hat{M}_\alpha \Psi' = \Psi \hat{W}_\alpha \hat{\Lambda}_\alpha \hat{V}_\alpha' \Psi'$. We can then consider the k th POP in physical space by making image plots of the back-transformed right singular vectors, $\Psi \hat{w}_{\alpha,k}$, for each $k = 1, \dots, p_\alpha$, and where $\hat{W}_\alpha \equiv (\hat{w}_{\alpha,1}, \dots, \hat{w}_{\alpha,p_\alpha})$.

For the monthly tropical Pacific SST anomaly data, consider the $p_\alpha = 20$ leading EOFs (which correspond to about 83% of the variation as shown in Figure 5.19). In this case, the decomposition of $\hat{M}_\alpha$ is summarized in Table 5.1.

Table 5.1 Eigenvalue Spectrum of $\hat{M}_\alpha$

$\hat{\lambda}_{\alpha,k}$	$\hat{\gamma}_{\alpha,k}$	$2\pi/\hat{\phi}_{\alpha,k}$
0.4529 + 0.0938i	0.4625	30.8
0.4529 - 0.0938i	0.4625	-30.8
0.6010 + 0.2146i	0.6381	18.3
0.6010 - 0.2146i	0.6381	-18.3
0.4703	0.4703	—
0.5027	0.5027	—
0.7246 + 0.2155i	0.7559	21.7
0.7246 - 0.2155i	0.7559	-21.7
0.9060	0.9060	—
0.8677 + 0.0798i	0.8713	68.5
0.8677 - 0.0798i	0.8713	-68.5
0.7223 + 0.1409i	0.7359	32.6
0.7223 - 0.1409i	0.7359	-32.6
0.8645	0.8645	—
0.7644 + 0.0600i	0.7668	80.2
0.7644 - 0.0600i	0.7668	-80.2
0.6583 + 0.0533i	0.6604	77.7
0.6583 - 0.0533i	0.6604	-77.7
0.7765	0.7765	—
0.6967	0.6967	—

We note from this table that all of the damping rates $\{\hat{\gamma}_{\alpha,k} : k = 1, \dots, 20\}$ are less than one, implying that there is no explosive growth potential in these data. We also note that there are many different oscillatory and non-oscillatory patterns present in the dataset. The oscillation frequencies vary quite a bit, with the estimated values of the associated periods, $2\pi/\hat{\phi}_{\alpha,k}$, falling between 18.3 to 80.2 months. As an example, consider the POP with estimated period of 68.5 months. The POP patterns associated with this oscillatory component are shown in Figure 5.23. The four panels in this figure show the POP evolution as indicated in (5.32). The associated POP-coefficient time series (real and imaginary components) are shown in Figure 5.24. We interpret Figures 5.23 and 5.24 as follows. The POPs show some similarity with the El Niño and La Niña patterns, and the time series in Figure 5.24 show increased amplitude during the major ENSO events. However, we note that the period of 68.5 months seems long for the average ENSO oscillation. Yet, there is nothing in the POP analysis to prevent a physical signal such as ENSO from being “distributed” among multiple normal modes. Examination of the other normal modes (not shown here) suggest that there are ENSO features found in multiple normal modes.

5.6 SPATIO-TEMPORAL CANONICAL CORRELATION ANALYSIS (CCA)

Canonical Correlation Analysis (CCA) is a long-standing multivariate statistical technique (Hotelling, 1936) that obtains linear combinations of two sets of random variables whose correlations are maximal. This approach has sometimes been used when the two random variables of interest are indexed by space and time. In this regard, it is related to multivariate EOFs, which are discussed in Section 5.4.2.

5.6.1 Two-Field Spatio-Temporal CCA

We assume that, in addition to the dataset $\{\mathbf{Z}_t \equiv (Z_t(s_1), \dots, Z_t(s_m))' : t = 1, \dots, T\}$, we are interested in another, related dataset $\{\mathbf{X}_t \equiv (X_t(\mathbf{x}_1), \dots, X_t(\mathbf{x}_\ell))' : t = 1, \dots, T\}$, with a possibly different spatial domain, but the same temporal domain. As before, we denote the covariance matrices by $\mathbf{C}_Z^{(0)} = \text{var}(\mathbf{Z}_t)$, $\mathbf{C}_X^{(0)} = \text{var}(\mathbf{X}_t)$, and $\mathbf{C}_{Z,X}^{(0)} = \text{cov}(\mathbf{Z}_t, \mathbf{X}_t)$, which have dimensions $m \times m$, $\ell \times \ell$, and $m \times \ell$, respectively. We then seek certain linear combinations of each of the data vectors, $a_t(k) = \xi_k' \mathbf{Z}_t$ and $b_t(k) = \psi_k' \mathbf{X}_t$, where $\xi_k \equiv (\xi_k(1), \dots, \xi_k(m))'$ and $\psi_k \equiv (\psi_k(1), \dots, \psi_k(\ell))'$. For $k = 1, 2, \dots, \min\{m, \ell\}$, define the k th *canonical correlation* as

$$r_k \equiv \text{corr}(\xi_k' \mathbf{Z}_t, \psi_k' \mathbf{X}_t) = \frac{\xi_k' \mathbf{C}_{Z,X}^{(0)} \psi_k}{(\xi_k' \mathbf{C}_Z^{(0)} \xi_k)^{1/2} (\psi_k' \mathbf{C}_X^{(0)} \psi_k)^{1/2}}. \quad (5.36)$$

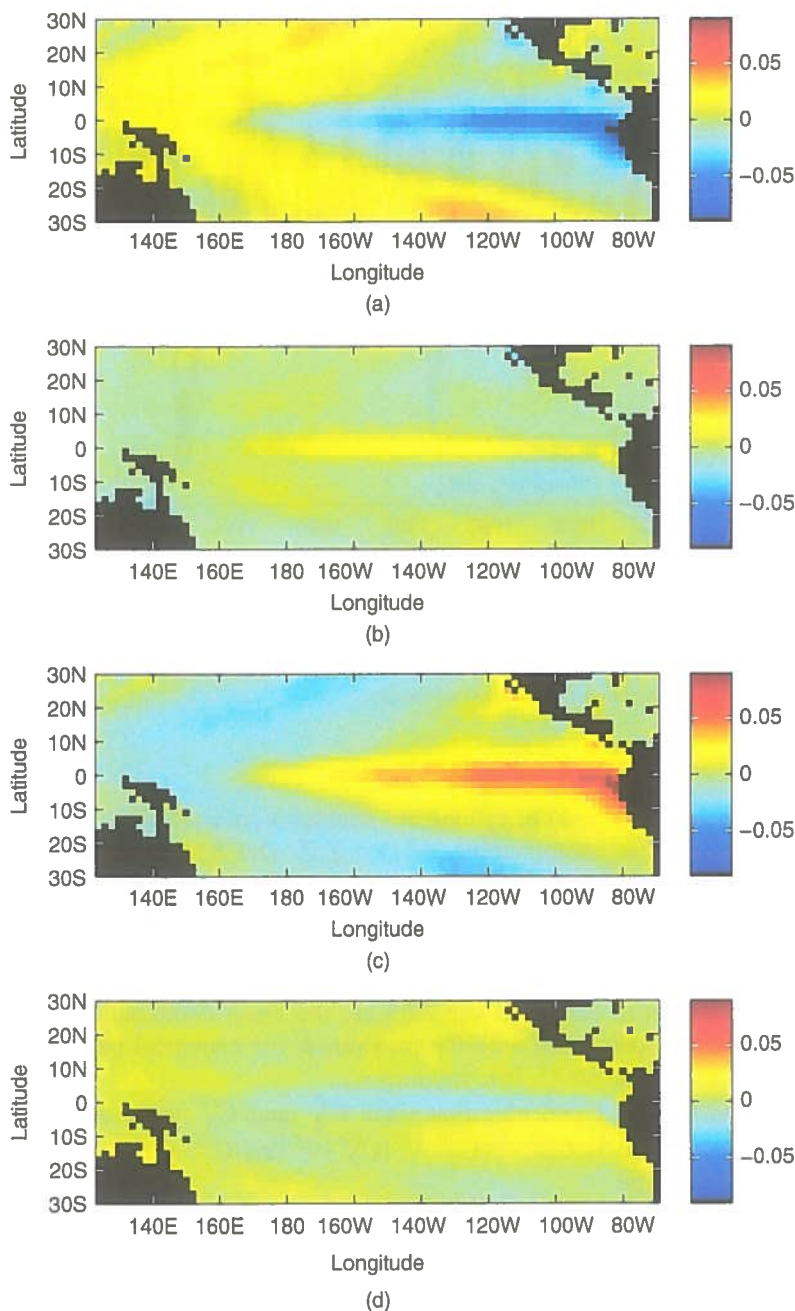


Figure 5.23 Image plots of the POP for the tropical monthly Pacific SST anomalies from January 1970 to December 2002. The POP corresponds to the 10th eigenvalue $\lambda_{\alpha,10} = 0.8677 + 0.0798i$, with damping rate 0.8713 and oscillation period 68.5 months, as shown in Table 5.1. (a) Real part ($\Psi \hat{\mathbf{w}}_{\alpha,10}^R$). (b) Negative imaginary part ($-\Psi \hat{\mathbf{w}}_{\alpha,10}^I$). (c) Negative real part ($-\Psi \hat{\mathbf{w}}_{\alpha,10}^R$). (d) Imaginary part ($\Psi \hat{\mathbf{w}}_{\alpha,10}^I$).

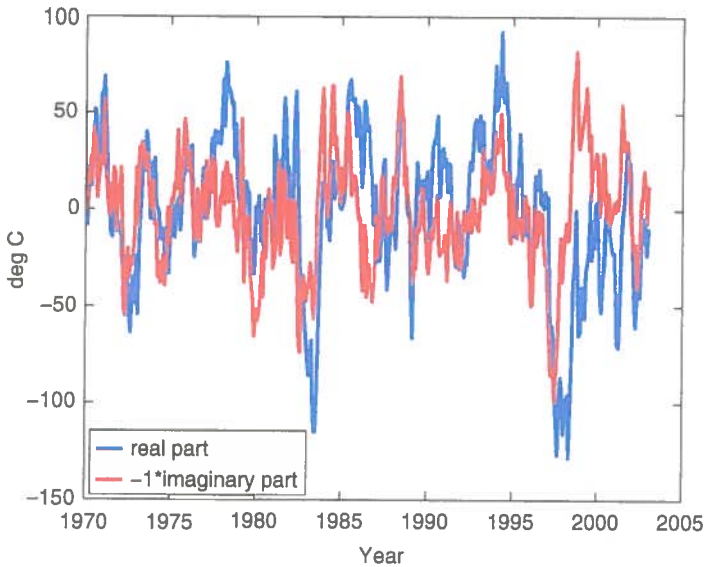


Figure 5.24 Monthly tropical Pacific SST anomaly data. Shown are the POP coefficient time series corresponding to the POPs in Figure 5.23. The real part of the POP coefficient time series is shown in blue and the negative of the imaginary part is shown in red. The y axis shows amplitude in units of $^{\circ}\text{C}$, and the x axis shows time in units of months.

The first pair (i.e., $k = 1$) of canonical variables associated with r_1 are defined as the set of linear combinations, $a_1(1) = \xi_1' \mathbf{Z}_t$ and $b_1(1) = \psi_1' \mathbf{X}_t$, such that $\text{var}(a_1(1)) = \text{var}(b_1(1)) = 1$, for $t = 1, \dots, T$, and ξ_1 and ψ_1 maximize the correlation (5.36), yielding r_1 . The second pair (i.e., $k = 2$) of canonical variables are then the linear combinations $a_2(t)$ and $b_2(t)$ that are uncorrelated with $a_1(t)$ and $b_1(t)$, have unit variance, and maximize (5.36), yielding r_2 . Then, the k th set of canonical variables are the linear combinations $a_t(k)$ and $b_t(k)$ that are uncorrelated with the previous $k - 1$ canonical pairs, have unit variance, and maximize (5.36), yielding r_k .

Initially, let $k = 1$ and note that since $\mathbf{C}_Z^{(0)}$ and $\mathbf{C}_X^{(0)}$ are positive-definite, they can be written as $\mathbf{C}_Z^{(0)} = (\mathbf{C}_Z^{(0)})^{1/2} (\mathbf{C}_Z^{(0)})^{1/2}$ and $\mathbf{C}_X^{(0)} = (\mathbf{C}_X^{(0)})^{1/2} (\mathbf{C}_X^{(0)})^{1/2}$. Then, we can write

$$r_1^2 = \frac{[\tilde{\xi}_1' (\mathbf{C}_Z^{(0)})^{-1/2} \mathbf{C}_{Z,X}^{(0)} (\mathbf{C}_X^{(0)})^{-1/2} \tilde{\psi}_1]^2}{(\tilde{\xi}_1' \tilde{\xi}_1)(\tilde{\psi}_1' \tilde{\psi}_1)}, \quad (5.37)$$

where $\tilde{\xi}_1 \equiv (\mathbf{C}_Z^{(0)})^{1/2} \xi_1$ and $\tilde{\psi}_1 \equiv (\mathbf{C}_X^{(0)})^{1/2} \psi_1$. The problem is now reduced to finding $\tilde{\xi}_1$ and $\tilde{\psi}_1$ that maximize (5.37). It can be shown (e.g., Johnson

and Wichern, 1992, p. 463) that r_1^2 is the largest singular value of the singular value decomposition of the cross-correlation matrix,

$$(\mathbf{C}_Z^{(0)})^{-1/2} \mathbf{C}_{Z,X}^{(0)} (\mathbf{C}_X^{(0)})^{-1/2}, \quad (5.38)$$

where $\tilde{\xi}_1$ and $\tilde{\psi}_1$ are the left and right singular vectors corresponding to r_1^2 , respectively. Consequently, we can obtain ξ_1 and ψ_1 through $\xi_1 \equiv (\mathbf{C}_Z^{(0)})^{-1/2} \tilde{\xi}_1$ and $\psi_1 \equiv (\mathbf{C}_X^{(0)})^{-1/2} \tilde{\psi}_1$, respectively. These are called the first CCA patterns, and they can be mapped on their respective regions. We can also obtain the time series of canonical variables, $a_t(1) \equiv \xi_1' \mathbf{Z}_t$ and $b_t(1) \equiv \psi_1' \mathbf{X}_t$, for $t = 1, \dots, T$.

In general, $\tilde{\xi}_k$ and $\tilde{\psi}_k$ are the left and right singular vectors, respectively, associated with the k th singular value r_k^2 from the singular value decomposition of (5.38). Then, for $k > 1$, ξ_k , ψ_k , $\{a_t(k)\}$, and $\{b_t(k)\}$ can be obtained analogously to the $k = 1$ case.

To calculate the singular value decomposition in (5.38), empirical covariance matrices, $\hat{\mathbf{C}}_Z^{(0)}$, $\hat{\mathbf{C}}_X^{(0)}$, and $\hat{\mathbf{C}}_{Z,X}^{(0)}$, are needed. These are given by (5.1) and (5.4), as discussed in Section 5.1.3. Then, the empirical singular values and singular vectors are obtained from the singular value decomposition of $(\hat{\mathbf{C}}_Z^{(0)})^{-1/2} \hat{\mathbf{C}}_{Z,X}^{(0)} (\hat{\mathbf{C}}_X^{(0)})^{-1/2}$. The CCA from these matrices is often unsatisfactory because the empirical covariance matrices are noisy (or singular) when calculated with sample sizes common in many spatio-temporal applications. To compensate for this, the data are often projected onto a truncated set of EOFs, just as in Section 5.5.1, before applying the singular value decomposition (e.g., von Storch and Zwiers, 1999, Chapter 4). As in Section 5.5.1 for POPs, we get a benefit from the reduction of spatial dimension; this is necessary for computational stability or if $\min\{m, \ell\} > T$ (otherwise, the method-of-moments estimate of (5.38) is undefined).

Example: To illustrate the exploratory application of CCA to spatio-temporal datasets, consider the tropical Pacific SST anomalies (Section 5.1), averaged over January through March (JFM) for each year from 1970 through 1999, as well as a dataset from the U.S. Fish and Wildlife Service Breeding Population Survey (BPS). Different from the BBS, the BPS is based on yearly aerial surveys that seek to assess the breeding population of waterfowl in North America. This survey has been conducted each year (since 1955) and consists of 18-mile linear segments (1/4 mile wide) over which the aircraft pilot and an observer count and speciate the waterfowl population. For this example, we consider “raw indicated pair count” data for Mallards from 1970 through 1999. The “indicated pair counts” consist of pairs (male and female) and “lone drakes” (individual males). Although there is clearly detection error in these counts, we ignore such error for this exploratory spatio-temporal data analysis. In addition, for simplicity of illustration, we only consider the ℓ BPS sampling locations for which there were no missing data for each of the 30 years from 1970 through 1999.

Figure 5.25 shows a spatial map of the empirical means and standard deviations of these counts at each of these sampling locations over the period of interest. We note that the largest average counts (and variability) occur in the center portion of the flyway. From a biological standpoint, one expects to see persistence in waterfowl breeding location as individuals tend to return to their natal landscape for breeding (i.e., “site philopatry”). However, there is inherent spatial variability in indicated pair counts since wetland and upland habitat

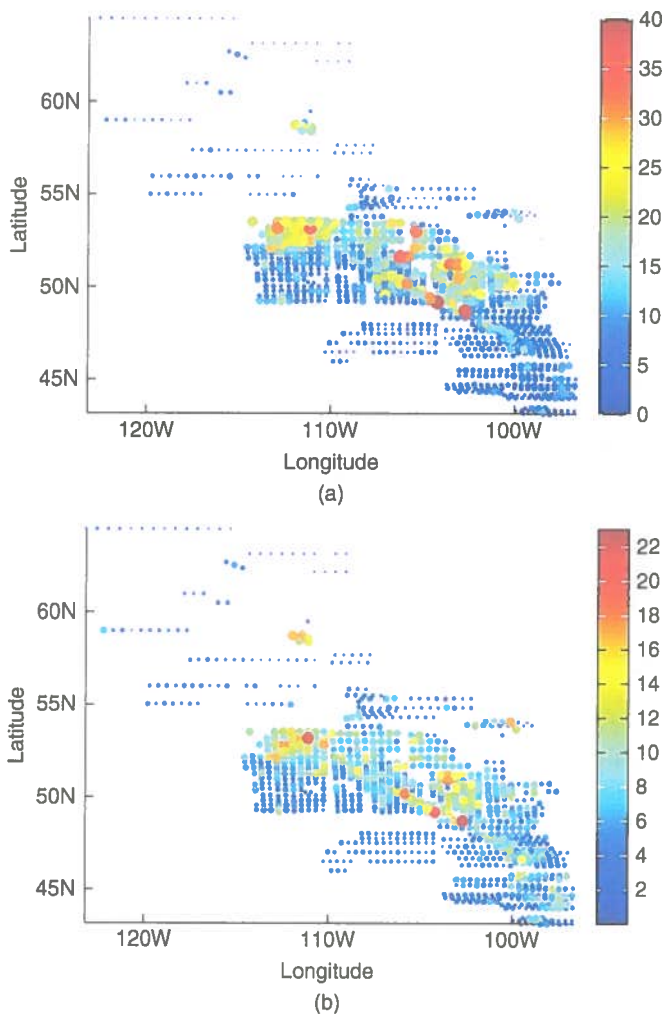


Figure 5.25 Summaries of indicated pair counts for Mallards, 1970–1999, obtained from the Breeding Population Survey; the region shown is the Prairie Pothole region of North America, and the color and radius of the circles are proportional to the size of each quantity plotted. (a) Averages taken over years. (b) Standard deviations taken over years.

conditions vary across the landscape. In addition, the waterfowl may fly over their natal landscape if the conditions are poor in any given year (e.g., in a drought), imposing temporal variability in breeding counts. Thus, one might hypothesize that since there is a strong connection between tropical Pacific SST conditions and the weather over North America (i.e., there is a *teleconnection* over large distances), there might be some evidence of a relationship between tropical Pacific SST anomalies and BPS counts. Therefore, we compare the average of the JFM SST anomalies from 1970 to 1999, to the BPS counts for the same period. The JFM period was chosen for the SST data, since the habitat conditions encountered by the migrating waterfowl would be determined from weather conditions in the late winter to early spring.

Figure 5.26 shows the first CCA patterns for both tropical Pacific SST anomalies and BPS counts. Figure 5.27 shows the time series associated with these patterns. These figures suggest that the relatively cooler ocean anomalies extending from the western Pacific up to the northern tropical Pacific (as well as cooler than normal SSTs off the coast of South America) are correlated with increased Mallard pair counts in the area centered on 110° W and between 50° and 55° N, as well as decreased counts in the area between 100° and 110° W and around 50° N. The two time series of canonical variables shown

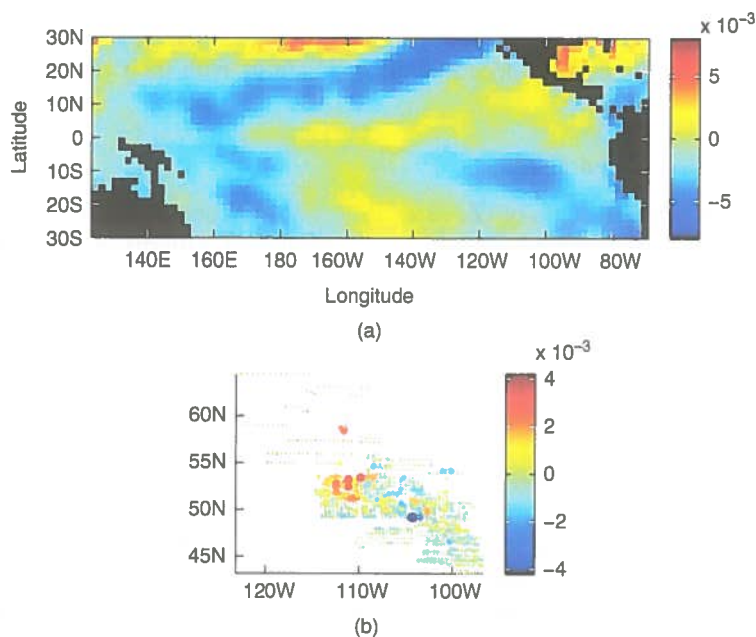


Figure 5.26 Yearly tropical Pacific SST anomaly data based on January–March averages, and yearly indicated pair counts for Mallards from the Breeding Population Survey. (a) First CCA pattern for SST anomalies. (b) First CCA pattern for the Mallard indicated pair counts. The CCA patterns are unitless.

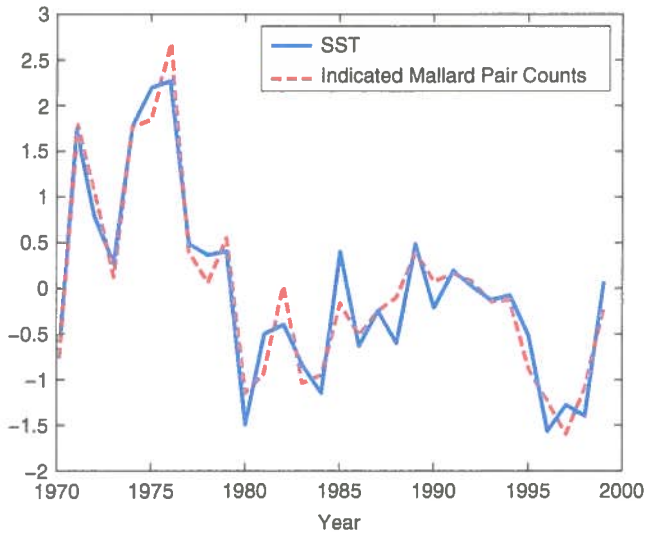


Figure 5.27 Time series of first canonical variables for yearly SST anomalies (blue solid line) and for yearly indicated pair counts for Mallards (red dashed line), associated with Figure 5.26. The x axis shows time in yearly increments. The time series are unitless (since CCA is based on correlations).

in Figure 5.27 show a very strong relationship (the canonical correlation is 0.96) and they both show decade-scale oscillations with their associated CCA patterns.

5.6.2 Time-Lagged CCA

When only one dataset (say $\{\mathbf{Z}_t : t = 1, \dots, T\}$) is considered, the CCA technique can be used to find the canonical correlation patterns between $\mathbf{Z}_t$ and $\mathbf{Z}_{t-\tau}$, for some time lag τ . In this case, we obtain the vectors ξ_k and ψ_k such that the marginal correlation between $a_t(k) \equiv \xi_k' \mathbf{Z}_t$ and $b_{t-\tau}(k) \equiv \psi_k' \mathbf{Z}_{t-\tau}$ is maximized. To do this, we obtain the k th left and right singular vectors from the singular value decomposition applied to $(\mathbf{C}_Z^{(0)})^{-1/2} \mathbf{C}_Z^{(\tau)} (\mathbf{C}_Z^{(0)})^{-1/2}$, where $\mathbf{C}_Z^{(\tau)}$ is the lag- τ cross-covariance matrix of $\{\mathbf{Z}_t\}$, which is assumed to be temporally stationary. Analogous to the two-field CCA previously described, to obtain ξ_k and ψ_k we have to weight the singular vectors obtained from this singular value decomposition by premultiplying them with $(\mathbf{C}_Z^{(0)})^{-1/2}$.

The time-lagged CCA outlined here is similar to the POP analysis discussed in Section 5.5. When considering POPs and time-lagged CCA (for $\tau = 1$), recall that the singular value decomposition must be performed on certain matrices, which we specify below.

$$\text{POP: } \mathbf{C}_Z^{(1)} (\mathbf{C}_Z^{(0)})^{-1}, \quad (5.39)$$

$$\text{CCA: } (\mathbf{C}_Z^{(0)})^{-1/2} \mathbf{C}_Z^{(1)} (\mathbf{C}_Z^{(0)})^{-1/2}. \quad (5.40)$$

Clearly, these two matrices are similar in that they include some form of the lag-one covariance matrix and the inverse of the lag-zero covariance matrix. It is straightforward to show that the singular values are the same for the decomposition of (5.39) and (5.40) and that the singular vectors from the CCA decomposition (5.40) are equivalent to those from the POP decomposition (5.39) scaled by the matrix $(\mathbf{C}_Z^{(0)})^{-1/2}$.

5.7 SPATIO-TEMPORAL FIELD COMPARISONS

There are numerous instances in spatio-temporal statistical analysis where there is strong interest in comparing two spatio-temporal “fields.” This might occur when comparing the output from two models (deterministic or statistical), comparing a historical forecast to the actual observations (or some smooth representation of them), or comparing model output/data for two or more time periods. For the purposes of this exposition, consider two spatio-temporal datasets represented by the $m \times T$ matrices, $\mathbf{Z} \equiv (\mathbf{Z}_1, \dots, \mathbf{Z}_T)$ and $\mathbf{X} \equiv (\mathbf{X}_1, \dots, \mathbf{X}_T)$.

There is a substantial literature on field comparison in the context of forecast verification in the atmospheric sciences (e.g., see the overview in Wilks, 2006, Chapter 7). Given that weather forecasters are often judged by the accuracy of their forecasts, there is a definite need for objective and quantitative forecast-verification methodologies to compare a forecasted field to the field that was subsequently observed. One such measure, commonly used by statisticians as well, is the *root mean squared error* (RMSE), which can be defined as (e.g., Briggs and Levine, 1997),

$$\text{RMSE} \equiv \frac{\|\mathbf{Z} - \mathbf{X}\|_F}{\sqrt{mT}},$$

where $\|\cdot\|_F$ is the Frobenius (i.e., Euclidean) norm. (The matrix $\mathbf{B} \equiv (b_{ij})$ has Frobenius norm, $\|\mathbf{B}\|_F \equiv \{\sum_i \sum_j b_{ij}^2\}^{1/2}$.)

Another commonly used measure is the *anomaly correlation coefficient* (ACC), which is the average gridpoint correlation between the anomalies from the two fields [where the anomalies are relative to some common underlying field, $\mathbf{C}$ (e.g., climatology)]. For example, Briggs and Levine (1997) write it as

$$\text{ACC} \equiv \frac{\text{tr}\{(\mathbf{Z} - \mathbf{C})'(\mathbf{X} - \mathbf{C})\}}{\|\mathbf{Z} - \mathbf{C}\|_F \|\mathbf{X} - \mathbf{C}\|_F}.$$

Such simple univariate summaries are not typically sufficient to describe the differences between fields. For example, the ACC measures the average linear association between the fields but does not account for the bias, whereas the RMSE does not take into account the linear association but is a good measure of average accuracy (Murphy, 1995; Briggs and Levine, 1997).

There are different approaches to obtaining more information about the fields. A classic analysis in this regard is the paper by Preisendorfer and Barnett (1983). In this case, they also consider the Frobenius norm between the two matrices but, more importantly, they show that this norm can be decomposed as the sum of three terms (i.e., “trinity statistics”) that relate to the difference in the centroids of the two spatio-temporal fields, their variability, and their evolution patterns. Significance tests can then be developed for these measures using a permutation procedure. One can also examine these measures across different spatial or temporal scales. For example, Branstator, Mai, and Baumhefner (1993) consider EOF-based diagnostics to compare fields by spatial scale. Similarly, Briggs and Levine (1997) use discrete wavelet transforms to decompose field verification measures. Direct comparison of *thresholded* fields through summary measures were given by Zhang, Craigmille, and Cressie (2008).

The methods described above provide overall summary statistics of the “difference” between two fields, but they are often unsatisfactory when comparing “spatially coherent” aspects. A spatial comparison of two fields was given by Shen, Huang, and Cressie (2002), who used the notion of false discovery rate in wavelet space to declare wavelet coefficients present or absent in the differenced field. In situations where a *feature* of one map (say a region of precipitation) is compared to another, traditional pixelwise comparisons may yield a poor match if the features are not aligned. It may be that just being able to capture the correct scale or orientation of the feature is suggestive of a “good” forecast, even if the spatial phase is not correct. That is, a forecast may have value even if a traditional field comparison suggests that the fields do not agree. For example, there may be a small location error causing the forecast and “truth” fields to barely overlap. In fact, the forecast might represent the size, shape, and intensity of the feature quite well, and a forecaster with expert knowledge could adjust the forecast to account for the uncertainty in location. Ebert and McBride (2000) describe a field-comparison procedure that differentiated location error from other sources of error. Micheas et al. (2007) extended methodology in morphometrics to field comparisons by using a Procrustes measure to decompose the “difference” in terms of location, scale, rotation angle, and intensity.

5.8 BIBLIOGRAPHIC NOTES

Visualization

The suggestions for data visualization presented in this section are quite simple, but they can be very useful. For example, the Hovmöller diagrams are quite helpful for visualizing propagating features. A more recent perspective on such diagrams can be found in Martius, Schwierz, and Davies (2006). In general, the visualization of statistical data is a topic that is constantly moving forward in response to new computational tools, software, exploratory techniques, and ever-more-complicated models. Chen, Hardle, and Unwin (2008)

and Carr and Pickle (2010) contain comprehensive and recent overviews of many of these methods. An older, but still quite useful, reference is Wegman and Carr (1993). The application and modification of these methods to the case of spatio-temporal data will be an exciting research topic in coming years.

Notice that the estimation of covariance matrices discussed in this section is covered in detail in almost any text that describes multivariate time series (e.g., Lütkepohl, 2005). In addition, estimation in the spatio-temporal geophysical context is discussed in von Storch and Zwiers (1999, Chapter 12).

Anselin (1995) defined the concept of local indicators of spatial association. A recent overview can be found in Lloyd (2007, Section 4.2.1). The foundations of visualizing multivariate data with parallel coordinates (PC) plots can be found in Inselberg (1985) and Wegman (1990). A recent example that considers spatio-temporal data can be found in Edsall (2003). A relatively recent overview of PC plots for spatial data can be found in Longley et al. (2005, Chapter 15).

Spectral Analysis

Cross-spectral analysis of time series is described in most advanced books on time-series analysis (e.g., Shumway and Stoffer, 2006; see also the Bibliographic Notes in Section 3.7). The data analysis by Wikle, Madden, and Chen (1997) illustrates how one can use a combination of cross-spectral analyses to make inference on propagating features in spatio-temporal datasets. Spatio-temporal spectral analysis (or *frequency–wavenumber analysis*) is summarized in von Storch and Zwiers (1999, Section 11.5).

EOFs

The definitive reference to EOFs is the monograph by Preisendorfer (1988). Additional general discussion from the geophysical perspective can be found in von Storch and Zwiers (1999) and Wilks (2006). The definitive monograph in Statistics that considers EOFs is Jolliffe (2002). In addition, any book on multivariate statistics (e.g., Johnson and Wichern, 1992) considers the closely related topic of principle components analysis.

In the meteorological literature, one finds extensive use of EOFs since their introduction by Lorenz (1956). For example, they have been used for describing climate (e.g., Kutzbach, 1967; Barnett, 1977), for comparing simulations of general circulation models (e.g., Preisendorfer and Barnett, 1983), for developing regression forecast techniques (e.g., Peagle and Haslam, 1982), in weather classification (e.g., Christensen and Bryson, 1966), in map typing (e.g., Richman, 1981), in the interpretation of geophysical fields (Obled and Creutin, 1986), and in the simulation of random fields, particularly nonhomogeneous processes (Braud and Obled, 1991). In addition, as in the psychometric literature for PCAs and as advocated by Richman (1981, 1986) and others, orthogonal and oblique rotation of EOFs often aids in the interpretation of spatio-temporal data (e.g., see Wilks, 2006, Section 11.5; Jolliffe, 2002, Chapter 11).

Because EOFs have difficulty resolving traveling-wave disturbances, complex EOF analysis was introduced by Wallace and Dickinson (1972) and was

shown to be very useful in applications to certain meteorological problems (e.g., Wallace, 1972; Barnett, 1983; Horel, 1984). For an overview from the geophysical perspective, see von Storch and Zwiers (1999, Chapter 16). From a statistical perspective, Brillinger (2001, Chapter 9) derives principal components in the frequency domain. We note that the notion of *extended EOF analysis* is related to so-called *singular spectrum analysis* (SSA), which considers principal components of an augmented data vector made up of lagged values from a univariate time series (e.g., Elsner and Tsonis, 1996; Jolliffe, 2002, Section 12.2.1). Indeed, in the case where one considers augmented vectors made up of lagged values from a multivariate dataset, then so-called *multichannel SSA* (MSSA) is equivalent to extended EOF analysis.

Although calculation of EOFs were obtained through the empirical covariance matrix, we note that it is equivalent to calculate EOFs through a singular value decomposition of the (centered) data matrix (e.g., von Storch and Zwiers, 1999, Section 13.2.9). In some software implementations, it may be more efficient to calculate them by this approach.

In cases where there is missing information or basis functions are needed for locations that are not observed, one can “shrink” the empirical covariance estimates to some parametric-based covariance matrix (e.g., Nott and Dunsmuir, 2002), or do likewise in the spectral domain (Daniels and Cressie, 2001). Alternatively, one can implement a preliminary smoothing procedure and then estimate the covariance matrix (e.g., Wikle and Cressie, 1999).

POPs

Principal Oscillation Patterns (POPs) were originally formulated as a specific case of Principal Interaction Patterns (PIPs) developed by Hasselmann (1988). POPs were reformulated by von Storch et al. (1988). Subsequently, there was a substantial increase in the number of POP applications in the literature and a comprehensive overview of POPs can be found in von Storch et al. (1995) and von Storch and Zwiers (1999). A recent example of the use of POPs in an ecological context to study the growth dynamics of shortleaf pine trees is given in Hooten and Wikle (2007).

The notion of complex POP (CPOP) analysis was introduced by Bürger (1993) as an extension of conventional POP analysis. Bürger notes that while POP analysis is able to model *traveling oscillations*, it is unable to model *standing oscillations*. In fact, he shows the inherent impossibility of modeling standing oscillations in first-order linear dynamical systems. Thus, CPOP analysis is a natural extension to POP analysis, in some ways analogous to the relationship between EOFs and CEOFs. That is, CEOFs are able to detect traveling disturbances, which cannot be detected by EOFs; and CPOPs can detect standing oscillations, which cannot be detected by POPs.

Traditional time-series-analysis techniques (including the vector-autoregression case that is analogous to POPs) require an assumption of second-order temporal stationarity (i.e., constant mean with autocorrelation depending only on time lag). This assumption clearly breaks down when the

physical process under consideration has known cycles (e.g., solar-influenced annual and semiannual cycles in atmospheric processes). In that case, the mean and variance (at least) are also periodic. Traditionally, these cycles are removed, and then it is hoped that the stationarity assumption is satisfied. Often, there is still evidence of nonstationarity, at least with regard to the variance. However, from a statistical perspective, it makes sense to use the redundant information contained in the periodically correlated statistics (i.e., cyclostationarity) optimally, rather than to remove it. An excellent discussion of the analysis of periodically correlated atmospheric time series can be found in Lund et al. (1995). Blumenthal (1991) first published the idea of using periodically correlated statistics in the POP framework; the approach is summarized in von Storch et al. (1995).

A more general decomposition that accounts for the variation in the data (EOFs) as well as the dynamics (POPs) is the *predictive oscillation pattern* methodology of Kooperberg and O'Sullivan (1996).

CCA

Canonical correlation analysis is described in detail in almost any text that covers multivariate statistics (e.g., Johnson and Wichern, 1992; Jolliffe, 2002). In the geophysical context, excellent overviews are contained in von Storch and Zwiers (1999) and Wilks (2006).

We note that Bretherton, Smith, and Wallace (1992) performed a comparison of methods for finding coupled patterns in climate data (including CCA and multivariate EOFs) and were supportive of a variant of CCA known (unfortunately) as Singular Value Decomposition (SVD). This is related to, but not the same as, the well known matrix-algebra technique of the same name. In this regard, von Storch and Zwiers (1999, Section 14.1.7) suggest that since this "SVD" approach effectively finds patterns that maximize the *covariance* between spatio-temporal fields, it is more appropriate to call this *maximum covariance analysis*. Some authors (e.g., Cherry, 1997) have pointed out that there are difficulties with the interpretation of CCA and "SVD" results, particularly the tendency for the methods to produce spurious spatial patterns. Nevertheless, the approach can be a useful exploratory tool with multivariate spatio-temporal data. In the atmospheric sciences, CCA has been used in diagnostic climatological studies (e.g., Glahn, 1968; Nicholls, 1987; Barnett and Preisendorfer, 1987), in the forecast of El Niño (Graham and Michaelsen, 1987; Barnston and Ropelewski, 1992), and in the forecast of long-range temperature and precipitation (Barnston, 1994).

The time-lagged CCA approach outlined here is similar to the POP analysis and to the minimum/maximum autocorrelation factor (MAF) analysis (Shapiro and Switzer, 1989; Cressie and Helderbrand, 1994; Solow, 1994). In the MAF case, one is interested in the eigenvectors proportional to those obtained from the singular value decomposition of $(\mathbf{C}_Z^{(0)})^{-1}\mathbf{V}$, where the matrix $\mathbf{V}$ is the first-difference covariance matrix given by $\mathbf{V} \equiv E[(\mathbf{Z}_t - \mathbf{Z}_{t-1})(\mathbf{Z}_t - \mathbf{Z}_{t-1})']$.

Field Comparisons

Spatio-temporal field comparison can borrow techniques and motivation from the forecast-verification literature (e.g., see von Storch and Zwiers, 1999, Chapter 18; Wilks, 2006, Chapter 7, and the references therein). Although a significant portion of that literature is related to categorical and probabilistic forecast *skill measures*, there is also a substantial component of the literature devoted to continuous-variable field comparisons. The methods of Preisendorfer and Barnett (1983) are particularly useful for spatio-temporal datasets. In addition, the review by Wigley and Santer (1990) contains many univariate and multivariate measures that are useful across a broad range of applications. We note that, given the statistical dependence usually found in spatio-temporal datasets, most of the methods of comparison rely on resampling or Monte Carlo tests (e.g., Stanford and Ziemke, 1994). Field comparisons based on “objects” have seen substantial development in the last few years (e.g., Hoffman et al., 1995; Ebert and McBride, 2000; Micheas et al., 2007). In addition, measures that explicitly account for distributional forecasts (or ensembles) are the subject of much interest (see Wilks, 2006, Section 7.7.1; Micheas et al., 2007).

A closely related area to field comparisons has developed out of what is known as “optimal fingerprint” analysis in the study of climate change (e.g., Hasselmann, 1979, 1993). It was developed for “distinguishing between the anticipated externally generated time-dependent climate-change signal and the natural internal variability of the climate system” (Hasselmann, 1993, p. 1957). This methodology has been extended into a Bayesian framework by Leroy (1998) and by Berliner, Levine, and Shea (2000).